

SODIUM BOROHYDRIDE IN CARBOXYLIC ACID MEDIA.

A REVIEW OF THE SYNTHETIC UTILITY OF ACYLOXYBOROHYDRIDES

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SODIUM BOROHYDRIDE IN CARBOXYLIC ACID MEDIA. A REVIEW

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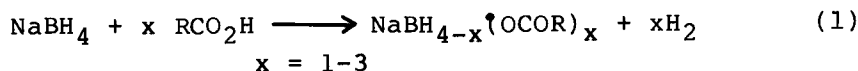
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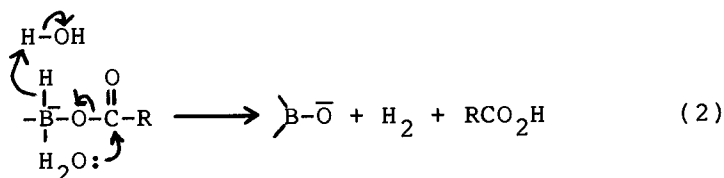
INTRODUCTION

Sodium borohydride (NaBH_4), alone or in conjunction with certain metals or solid supports, is one of the most useful reagents in chemistry.¹ Modified versions of NaBH_4 , such as sodium cyanoborohydride (NaBH_3CN), also have widespread utility in synthesis.² A relatively new modified- NaBH_4 reagent is that produced when NaBH_4 is allowed to react with a carboxylic acid (RCO_2H) (Eq. 1). The resulting sodium acyloxyborohydrides and their use in organic synthesis are the subject of this review.³

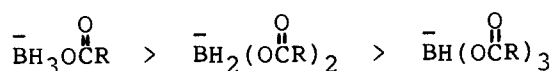


Unlike the reaction of NaBH_4 with mineral acids or aqueous acids,⁴ which leads to diborane formation or complete hydrolysis, the reaction of NaBH_4 with neat carboxylic acids (RCO_2H) or solutions of RCO_2H in nonprotic solvents leads to the formation of acyloxyborohydrides. Depending on the relative concentration of RCO_2H , one, two, or three hydrides will be available for reaction. Indeed, as will be seen, even in the presence of excess RCO_2H the triacyloxyborohydride species ($x = 3$, Eq. 1) is relatively stable and only surrenders its

last hydride upon heating or prolonged exposure to RCO_2H . However, all three types of acyloxyborohydrides are rapidly hydrolyzed by water (Eq. 2).



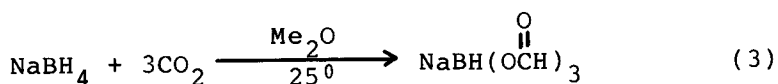
As will be apparent in this review, the fact that one can in principle control the number and kind of acyloxy groups on the boron atom leads to remarkable chemoselectivity. The data thus far accumulated indicate the following order of decreasing hydride-donating ability.



This reactivity order is presumably a consequence of both the inductive electron-withdrawing ability of the acyloxy group (e.g., $\sigma_I = 0.39$ for OAc)⁵ which strengthens the B-H bond and the steric bulk surrounding the B-H bond.

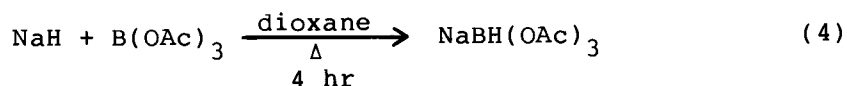
I. HISTORICAL. DISCOVERY AND CHARACTERIZATION OF ACYLOXYBOROHYDRIDES

Interestingly, the first reported synthesis of an acyloxyborohydride, in 1955, did not involve carboxylic acids. Wartik and Pearson⁶ prepared sodium triformyloxyborohydride by allowing NaBH_4 to react with carbon dioxide in dimethyl ether at room temperature (Eq. 3).



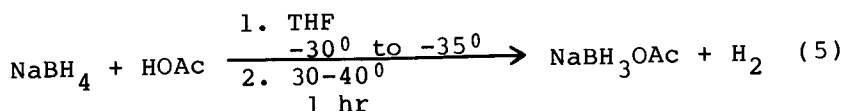
These workers noted that $\text{NaBH}(\text{OCHO})_3$ reacts rapidly with dilute aqueous acid to give dihydrogen, formic acid, and boric acid in the expected stoichiometry. Moreover, they made the important observation that $\text{NaBH}(\text{OCHO})_3$ decomposes on standing, or more rapidly on melting, to give methyl formate. This result implicates the formation of methanol by the self-reduction of $\text{NaBH}(\text{OCHO})_3$ to formaldehyde, thence to methanol, and finally to methyl formate; the ramifications of this observation will be seen later.

At about the same time, Nenitzescu and Badea⁷ reported the synthesis of $\text{NaBH}(\text{OAc})_3$, as "a white solid, insoluble in organic solvents," from the reaction of $\text{B}(\text{OAc})_3$ and sodium hydride in boiling dioxane (Eq. 4). A small amount of $\text{NaBH}_2(\text{OAc})_2$ was reported to be present in the filtrate from which $\text{NaBH}(\text{OAc})_3$ precipitated. These workers also noted that $\text{NaBH}(\text{OAc})_3$ decomposes in moist air and in water. Two years later Reetz⁸ and Brown and Subba Rao⁹

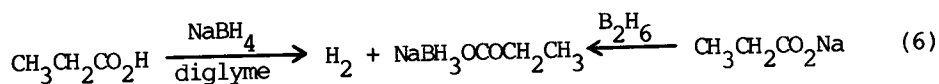


independently described the formation of acyloxyborohydrides from the reaction of NaBH_4 with RCO_2H (Eqs. 5 and 6).

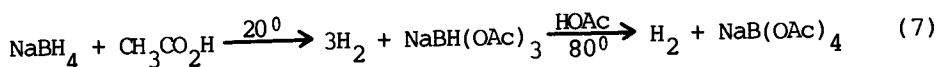
Reetz⁸ isolated NaBH_3OAc from NaBH_4 and acetic acid in tetrahydrofuran (THF) (Eq. 5), and provided some analytical data in support of the structure. Thus, on reaction with water this substance liberates three moles of dihydrogen. Moreover, no diborane can be detected on heating $\text{NaBH}_3\text{OCOCH}_3$ at 55° for 10 min, although it does react with trialkylphosphites to form $(\text{RO})_3\text{PBH}_3$ in good yield.



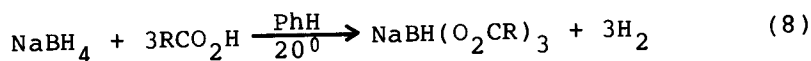
Brown and Subba Rao⁹ proposed the formation of the related propionic acid derivative (Eq. 6) but no experimental evidence was advanced to support its structure. They also suggested that the reaction of diborane with sodium propionate led to the same material.



Several years later, we^{10,11} and Marchini *et al.*¹² observed that NaBH_4 reacts with excess glacial acetic acid to liberate 3 moles of dihydrogen (Eq. 7). The last hydride is released slowly at 20° or more rapidly on heating or in the presence of water.



Marchini and coworkers¹² also reported the preparation and chemical, physical, and spectral properties of several acyloxyborohydrides (Table 1), prepared according to Eq. 8;



they also observed that $\text{NaBH}(\text{OCOC}_6\text{H}_5)_3$ undergoes self-reduction in refluxing toluene to give benzyl alcohol.

Another Italian group¹³ prepared sodium tris(trifluoroacetoxy)borohydride (Eq. 9) and observed a mp of 64-66° and bands at 1775 and 1680 cm^{-1} in the infrared spectrum.

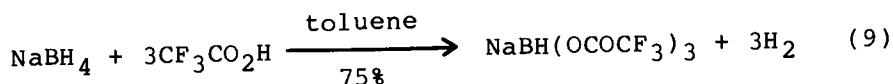
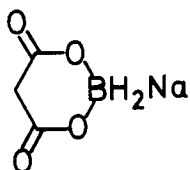


TABLE 1. Properties of Sodium Triacyloxyborohydrides¹²

Compound	mp (°C)	IR (cm ⁻¹)	
		B-H	C=O
NaBH(OCHO) ₃	>300°	2480	1680
NaBH(OAc) ₃	>300°	2480	1660
NaBH(OCOPh) ₃	>300°	2490	1670, 1635
NaBH(OCOCH ₂ Cl) ₃	120-5° (dec)	2530	1735, 1685

Egan and Morse¹⁴ have recorded the IR spectrum of NaBH₃OAc and observed 2500 and 1683 cm⁻¹ for the B-H and C=O stretching absorptions, respectively. These workers also noted, as did Hui,¹⁵ that NaBH₂(OAc)₂ could not be prepared cleanly.

However, Hui¹⁵ was able to synthesize the malonic acid-derived acyloxyborohydride shown below, perhaps the only known stable diacyloxyborohydride species.



The remaining few cases of acyloxyborohydride isolation and study will be presented in the appropriate section to follow.

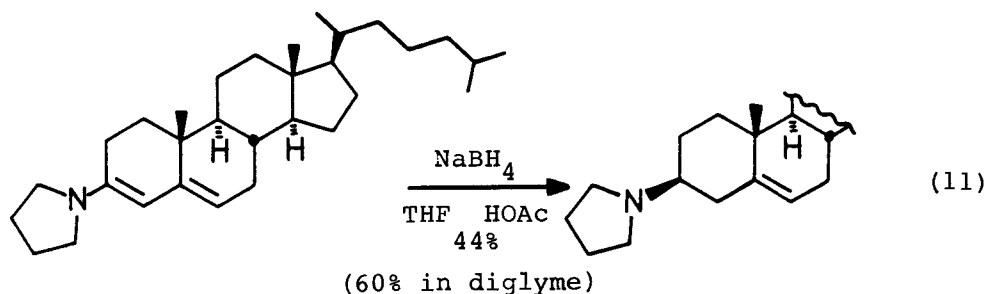
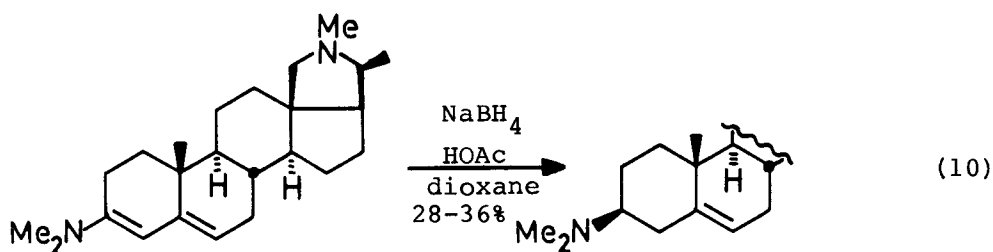
In most of the examples of the use of acyloxyborohydrides in synthesis (vide infra), the reagent is not isolated per se but, rather, is generated and utilized in situ. Therefore, in the ensuing discussion we have not specified the actual acyl-

oxyborohydride reagent, except where it has been isolated and employed as such.

Finally, it will be noted that this review covers also NaBH_3CN , LiBH_4 , KBH_4 , and $n\text{-Bu}_4\text{NBH}_4$ in combination with carboxylic acids.

II. REDUCTION OF ENAMINES

Apparently, the first reported use of $\text{NaBH}_4/\text{RCO}_2\text{H}$ in organic synthesis was the reduction of two steroidal di-enamines by Marshall and Johnson¹⁶ (Eq. 10 and 11) and, in fact, was the final step in their total synthesis of ($\pm$)-conessine^{16c} (Eq. 10).

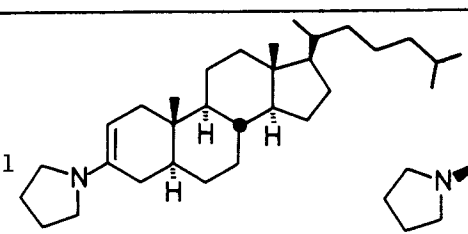
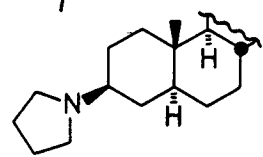
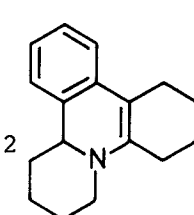
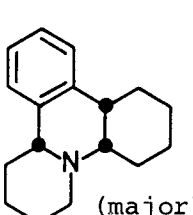
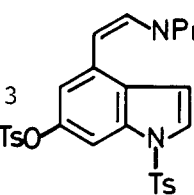
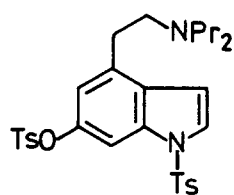
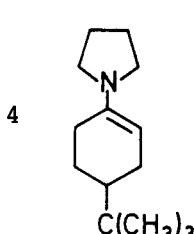
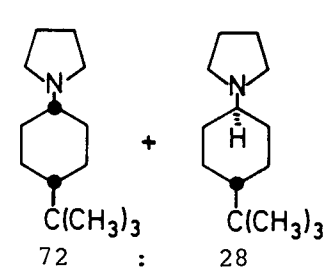


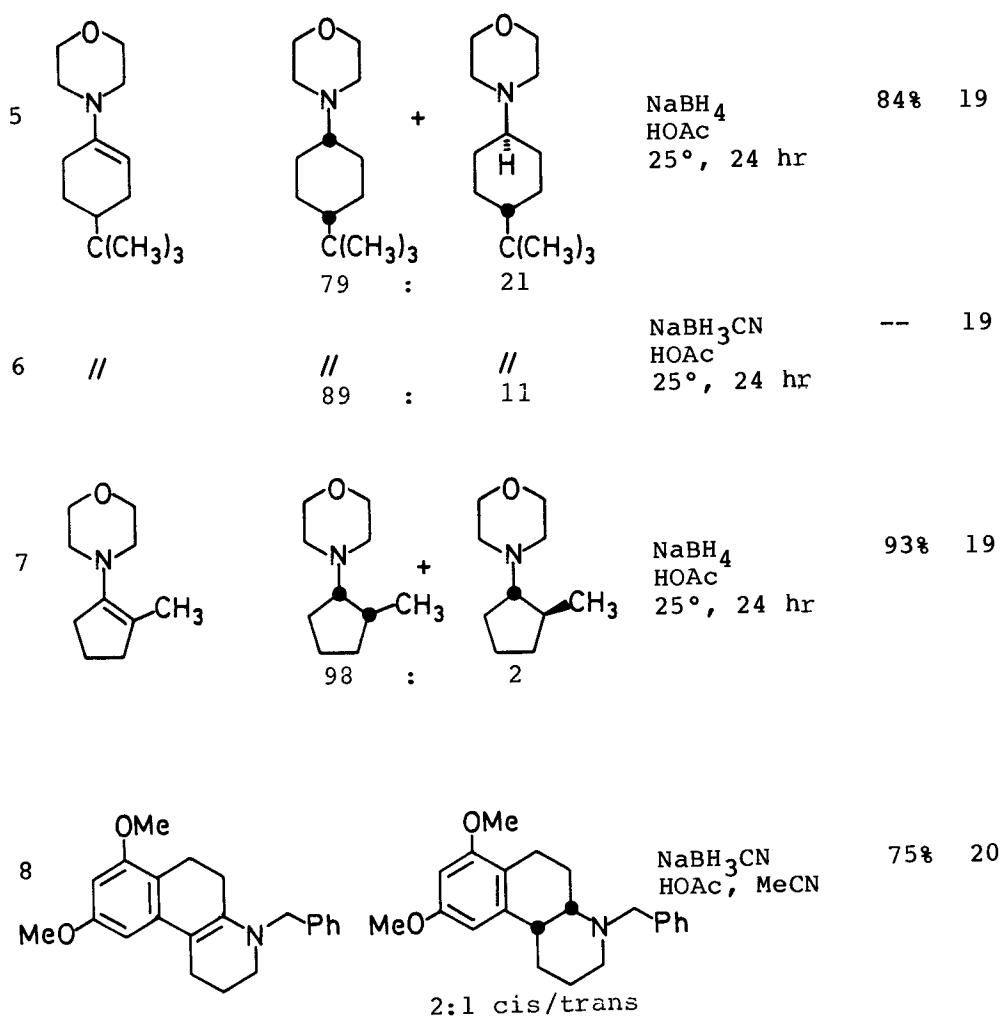
These workers also showed that simple enamines were reduced under these conditions, and since then a number of other enamine reductions have been described (Table 2). Noteworthy is the extensive study by Hutchins¹⁹ (entries 4-7). Sodium cyanoborohydride can be substituted for NaBH_4

(entries 3, 6, 8), especially if amine alkylation is to be avoided (vide infra).

From these studies it is clear that reductions of en-
amines (via immonium ions) with sodium triacetoxymborohydride
are reasonably (entries 2, 4-6) to highly (entry 7) stereo-
selective, with the preferred approach being from the less
hindered side (equatorial attack) to give the axial product.

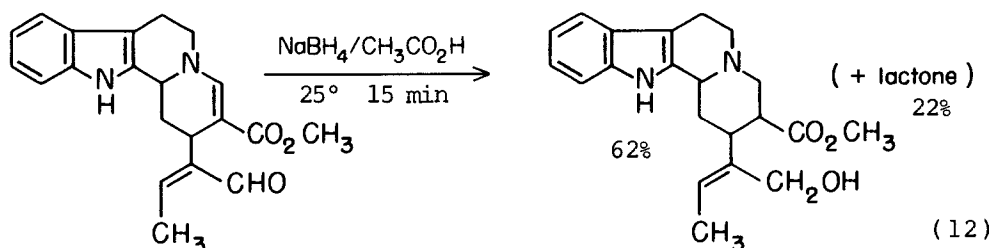
TABLE 2. Reduction of Enamines

Entry	Substrate	Product	Conditions	Yield	Ref.
1			NaBH ₄ , THF HOAc, Δ, 1 hr	70%	16a
2		 (major)	NaBH ₄ , THF HOAc, rt	80%	17
3			NaBH ₃ CN MeCN HOAc rt, 10 min	78%	18
4		 72 : 28	NaBH ₄ HOAc 25°, 24 hr	78%	19



III. REDUCTION OF VINYLOGOUS AMIDES, CARBAMATES, UREAS AND N-ACYLENAMINES

Another pioneering application of $\text{NaBH}_4/\text{RCO}_2\text{H}$ methodology was the chemoselective reduction of the vinylogous carbamate double bond in vallesiachotamine as reported in 1966 by Djerassi²¹ (Eq. 12). The aldehyde functionality was also reduced, but, interestingly, the indole double bond was not reduced, in contrast to studies discussed later (Section V).

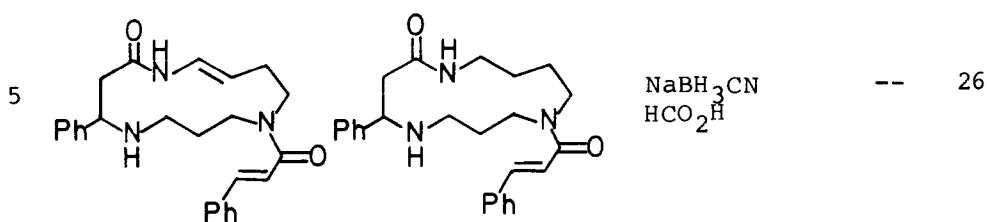


Several other examples of this particular reduction have been revealed (Table 3). Noteworthy is the fact that the stronger trifluoroacetic acid can be used (entry 3) and that acyclic systems may undergo β -elimination (entry 4).

TABLE 3. Reduction of Vinylogous Amides, Carbamates, Ureas and N-Acylenamines

Entry	Substrate	Product	Conditions	Yield	Ref.
1			NaBH ₄ HOAc 0°→rt	80%	22
2			NaBH ₃ CN HOAc THF MeOH 45°, 4 h	75%	23
3			NaBH ₄ CF ₃ CO ₂ H PhH	55%	24
4			NaBH ₄ CF ₃ CO ₂ H <u>i</u> -PrOH	60-80%	25

R₁ = i-Pr, Ph, C₆H₁₁
 R₂ = i-bu, Pr, neo-pentyl
 R₃ = Et, -CH₂CH₂OCH₂CH₂-

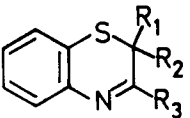
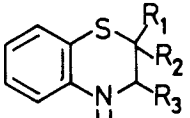
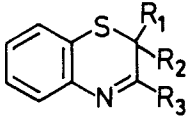
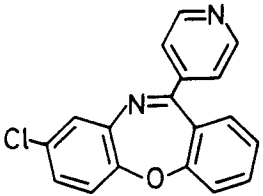
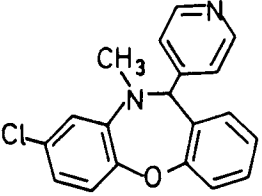
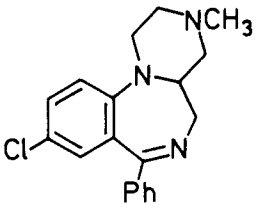
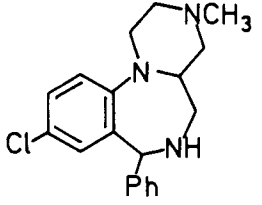
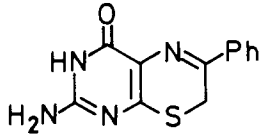
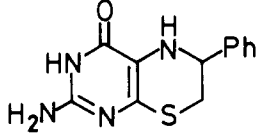


IV. REDUCTION OF IMINES, IMMONIUM SALTS, AND RELATED SYSTEMS

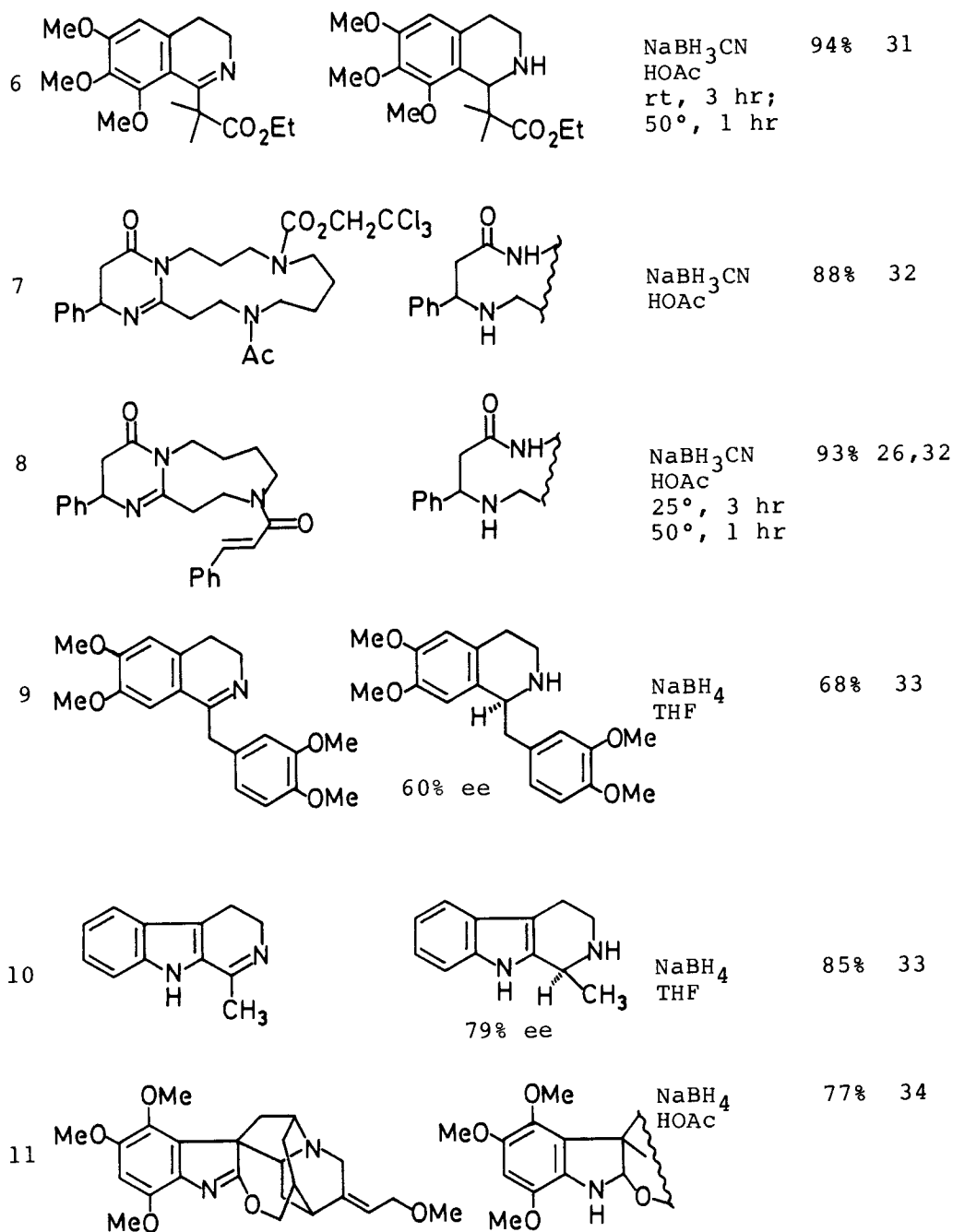
In view of the results described in the previous two sections, it is not surprising that imines and immonium salts are smoothly reduced to amines (Table 4). Depending on the system and reaction conditions, the initially-produced amine may be N-alkylated by the carboxylic acid (entries 2, 3, 13). This novel amine alkylation will be discussed in detail in Section VI. The NaBH₄/RCO₂H reduction of imines is analogous to that utilizing NaBH₃CN/MeOH/pH 3.²⁷ Several points about Table 4 should be made. Pyridine, pyrimidine, and furan rings (entries 3, 5, 15, 16) are generally inert to the action of NaBH₄ (or NaBH₃CN)/RCO₂H. In some cases very useful ring cleavage is observed (entries 7, 8, 12) and Wasserman has made extensive use of this reductive cleavage in his elegant syntheses of spermine/spermidine alkaloids.^{26,32} The N-trifluoroethylation (entry 13) can be suppressed by using NaBH₃CN/CF₃CO₂H. It is interesting to note that the acetic acid-induced ring opening observed by Sakai³⁴ (entry 12) is not observed when trifluoroacetic acid is used (entry 13). Especially noteworthy is the high degree of asymmetric reduction observed with a NaBH₄/proline acyloxyborohydride complex (entries 9, 10). Several other imines and optically active amino acids were examined in this important study.³³ Finally, Weinreb has used this methodology (NaBH₃CN/CF₃CO₂H) to effect a

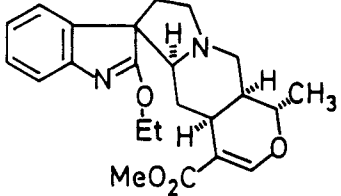
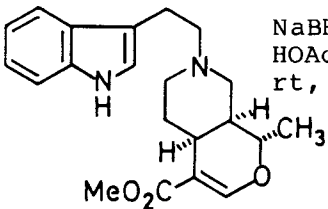
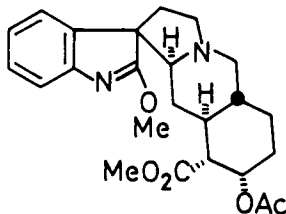
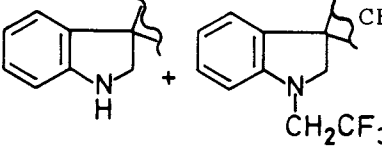
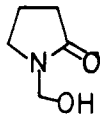
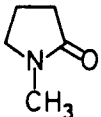
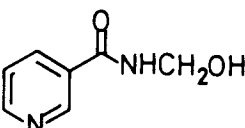
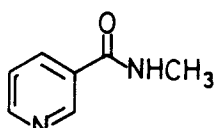
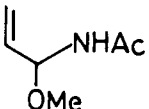
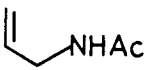
convenient N-methylation of primary and secondary amides (entries 14-16), in a transformation that presumably involves the formation and subsequent reduction of acylimmonium ion intermediates (cf. Table 3, entry 5 for a related example).

TABLE 4. Reduction of Imines, Immonium Salts, and Related Systems

Entry	Substrate	Product	Conditions	Yield	Ref.
1	 a; $R_1 = R_2 = \text{Me}$, $R_3 = \text{Ph}$ b; $R_1 = \text{H}$, $R_2 = R_3 = \text{Ph}$		NaBH_4 (1 eq.) $\text{CH}_3\text{CO}_2\text{H}$ 80° , 1 hr	60-95%	12
2	 a; $R_1 = R_2 = \text{Me}$, $R_3 = \text{Ph}$ b; $R_1 = \text{H}$, $R_2 = R_3 = -(\text{CH}_2)_5-$		NaBH_4 (5 eq.) HOAc 80° , 3 hr	75-95%	12
3	 (others)		NaBH_4 HCO_2H $10^\circ \rightarrow 25^\circ$	84%	28
4			NaBH_3CN HOAc rt	95%	29
5			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ rt	51%	30

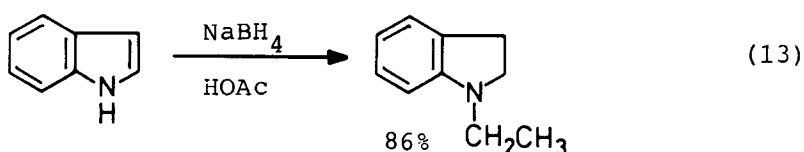
GRIBBLE AND NUTAITIS



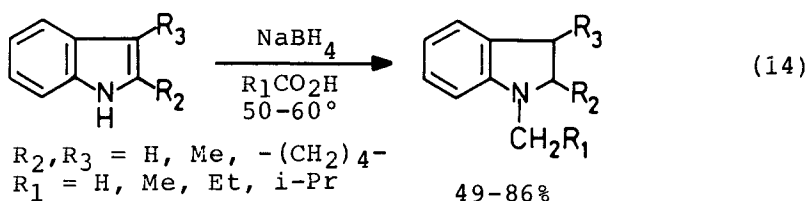
12			NaBH ₄ HOAc rt, 3 hr	55%	34
13			NaBH ₄ CF ₃ CO ₂ H	--	35
14			NaBH ₃ CN CF ₃ CO ₂ H rt, 1-2 days	90%	36a
15			NaBH ₃ CN CF ₃ CO ₂ H rt, 1-2 days	83%	36a
16			NaBH ₃ CN CF ₃ CO ₂ H rt, 1-2 days	68%	36a
17			NaBH ₄ CF ₃ CO ₂ H PhH rt, 10 min	--	36b

V. REDUCTION OF INDOLES

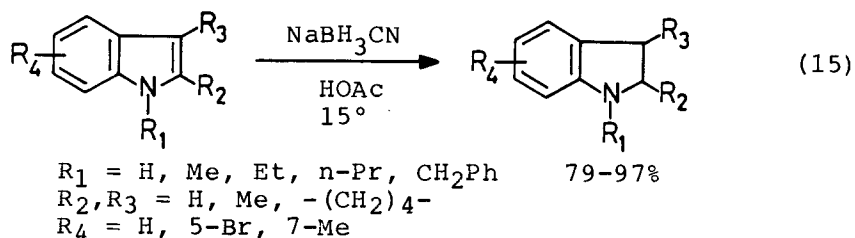
Our own research in the area of NaBH₄/RCO₂H methodology began in 1973 when the senior author attempted to reduce indole to indoline with NaBH₄ in glacial acetic acid.³⁷ Much to our surprise, the product was not indoline but rather N-ethylindoline in 86% distilled yield! (Eq. 13).



Control experiments and other data show that indoline is formed rapidly and then undergoes N-alkylation to give product. Details of this N-alkylation will be described in Section VI. This synthesis of N-alkylindolines is general for a variety of indoles and carboxylic acids (Eq. 14).¹¹



By using NaBH₃CN in place of NaBH₄, one can avoid N-alkylation and achieve a very simple and efficient synthesis of indolines (Eq. 15).³⁸ Only those indoles having electron-withdrawing groups fail to undergo reduction (e.g., 5-nitro- and 2,3-diphenylindole); this modification using NaBH₃CN/HOAc to reduce indoles to indolines was recently "rediscovered" by Kumar and Florvall.³⁹



It is important to note that earlier workers did not observe reduction of the indole double bond because in these systems (e.g. Eq. 12; Table 3, entries 1, 2; Table 4, entry 12) a basic nitrogen atom is present which, when protonated, protects the indole ring from protonation and reduction. However, as can be seen in Table 5, the stronger trifluoroacetic acid overcomes this difficulty and reduction of the indole double bond can be achieved.

Several additional examples of the use of NaBH_4 (or NaBH_3CN)/ RCO_2H to reduce the indole ring are tabulated in Table 5. A striking example of the inherent chemoselectivity noted earlier is seen in the reduction of only the more basic indole double bond in the molecule shown in entry 3.

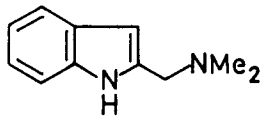
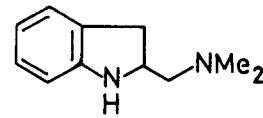
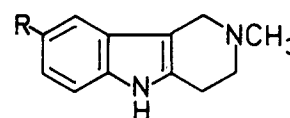
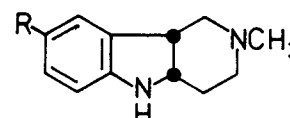
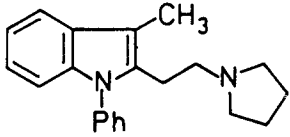
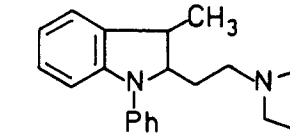
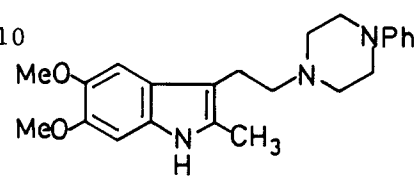
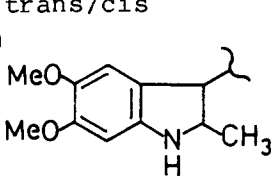
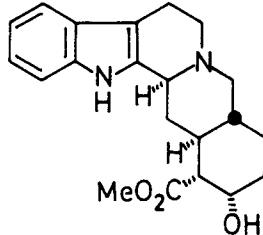
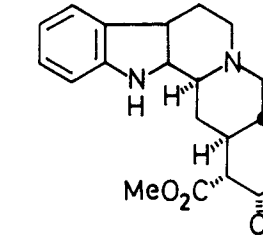
Generally, the use of trifluoroacetic acid does not give much N-trifluoroethylation; however, if this becomes a problem (entry 6), then NaBH_3CN can be substituted for NaBH_4 . Alternatively, lesser amounts of NaBH_4 (or KBH_4) may be used (entry 8).

As discussed in Section II, the reduction can be highly stereoselective giving product resulting from axial protonation and hydride delivery from an equatorial direction (entries 4, 5, 8). However, in simple indole systems there is virtually no stereoselectivity (entries 9, 10).

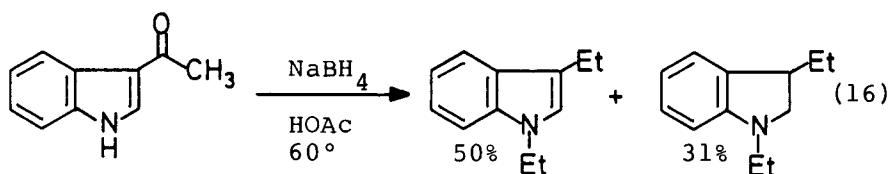
Indole itself undergoes an interesting reaction with $\text{NaBH}_4/\text{CF}_3\text{CO}_2\text{H}$, which will be discussed in Section XIX.

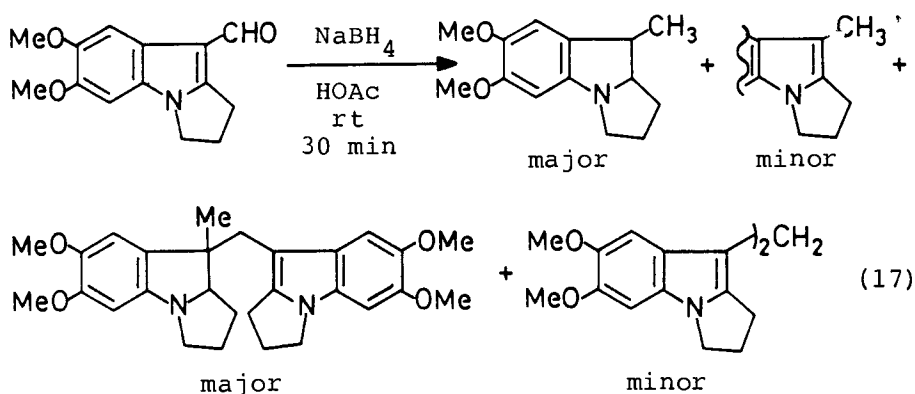
TABLE 5. Reduction of Indoles

Entry	Substrate	Product	Conditions	Yield	Ref.
1			NaBH ₃ CN HOAc rt	86%	40
2			NaBH ₄ HOAc rt, 30 min	100%	41
3			NaBH ₃ CN HOAc rt	78%	42
4			NaBH ₄ CF ₃ CO ₂ H 0°	90%	11,43
5			NaBH ₃ CN MeOH CF ₃ CO ₂ H	82%	44
6			NaBH ₃ CN CF ₃ CO ₂ H (NaBH ₄ CF ₃ CO ₂ H)	81% (35%)	45

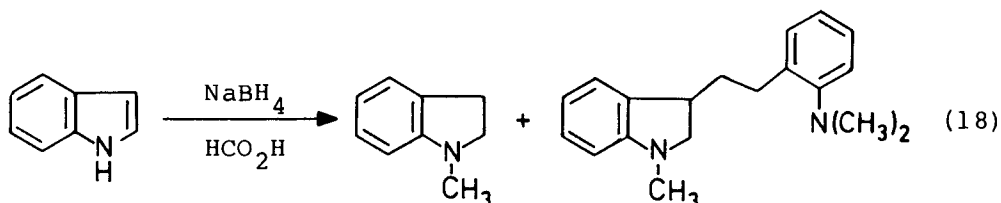
7			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ $0^\circ-10^\circ$	67%	46
8			1-2 moles KBH_4 (or NaBH_4) $\text{CF}_3\text{CO}_2\text{H}$	80%	47
R = Me, Cl, F, Br, CO_2Et					
9			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$	>95% (gc)	48
60:40 trans/cis					
10			NaBH_3CN $\text{CF}_3\text{CO}_2\text{H}$	84%	49
56:44 trans/cis					
11			NaBH_4 (or NaBH_3CN) $\text{CF}_3\text{CO}_2\text{H}$	80-100%	35,50

The reaction of 3-acylindoles with $\text{NaBH}_4/\text{RCO}_2\text{H}$ can take a complicated course (Eq. 16⁵¹ and 17⁴¹).





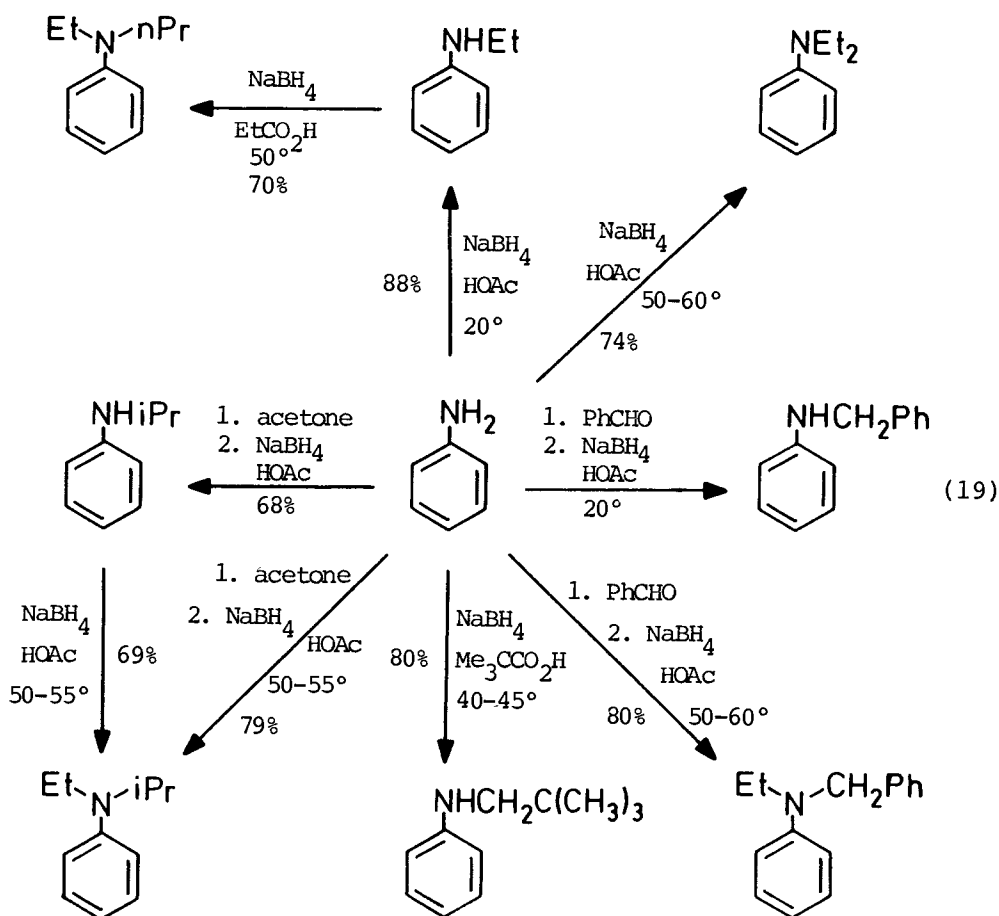
Likewise, the reaction of indole with $\text{NaBH}_4/\text{HCO}_2\text{H}$ gives, in addition to the expected N-methylindoline (Eq. 14),¹¹ the dimeric product shown in Eq. 18.⁵² This aberrant pathway has not been observed with other carboxylic acids.



VI. N-ALKYLATION OF AMINES

Perhaps the most extraordinary property of the $\text{NaBH}_4/\text{RCO}_2\text{H}$ reagent is its ability to N-alkylate amines, alluded to several times earlier. We believe that the mechanism for this transformation involves self-reduction of the acyloxyborohydride species to give free aldehyde (or its synthetic equivalent) followed by condensation with the amine and reduction to the N-alkylated amine.¹¹

The power and versatility of this amine alkylation methodology is illustrated in Eq. 19.^{11,51}

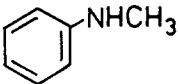
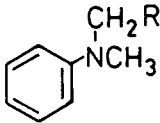
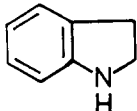
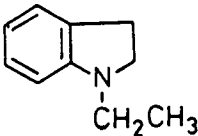
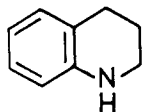
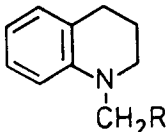


Thus, one can prepare unsymmetrical tertiary amines from primary amines in one pot, introduce the very bulky neopentyl group using pivalic acid, control the reaction (in some cases) so as to stop at the secondary amine stage, and use ketones so as to introduce secondary alkyl groups.

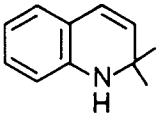
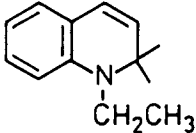
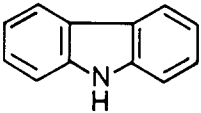
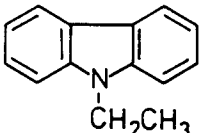
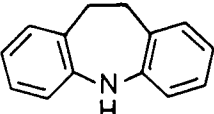
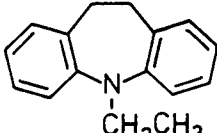
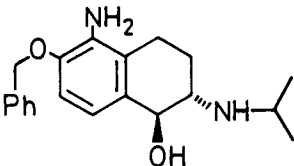
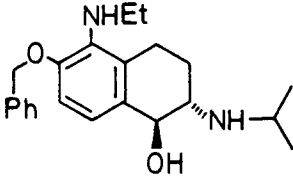
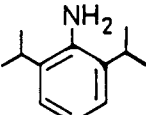
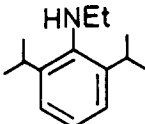
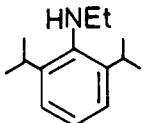
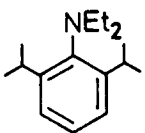
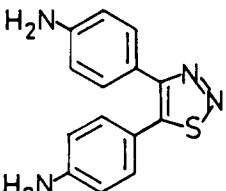
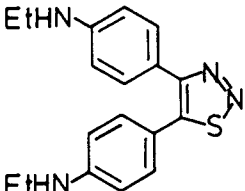
Simultaneously and independently of our own work,¹¹ Marchini and coworkers¹² also discovered this amine N-alkylation and extended it to the use of solid carboxylic acids in cosolvents. This important contribution as well as other examples of this amine N-alkylation are tabulated in Tables 6 (aromatic amines) and 7 (aliphatic amines).

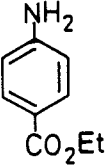
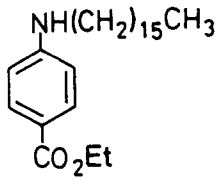
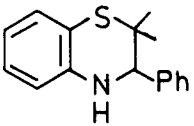
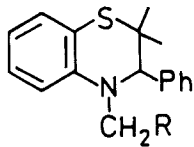
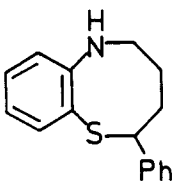
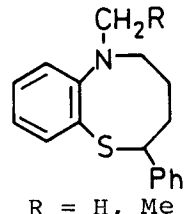
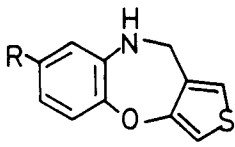
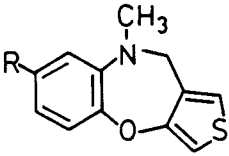
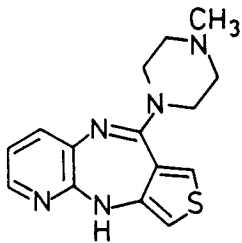
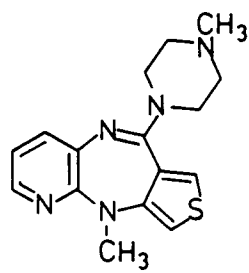
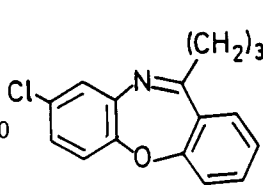
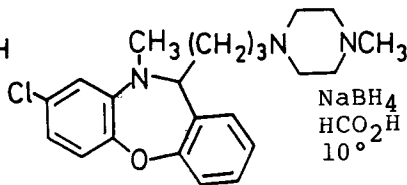
The N-alkylation of aromatic amines works equally well for nonbasic amines (entries 6-9), can be made chemoselective in the presence of an aliphatic amine (which requires higher temperatures for N-alkylation, cf., Table 7) (entries 10, 11, 23), and can be controlled so as to give mono- or dialkylation (entries 10-14). Moreover, a variety of functional groups (hydroxyl, alkene, carboethoxy, sulfur, amide, aryl ketone) and heterocyclic rings (pyridine, thiophene, thiadiazole) are unaffected by the appropriate $\text{NaBH}_4/\text{RCO}_2\text{H}$ conditions. In contrast to other carboxylic acids, trifluoroacetic acid gives lower yields of N-trifluoroethylation in most cases (entries 21-25).

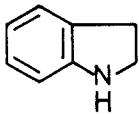
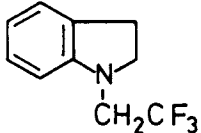
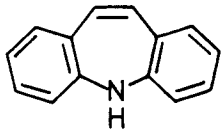
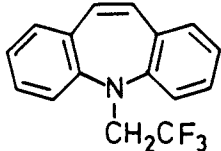
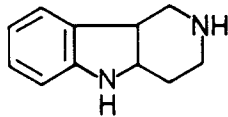
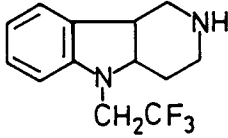
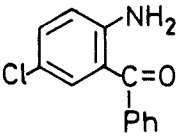
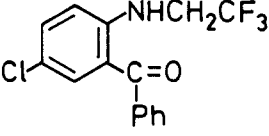
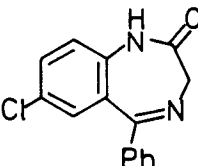
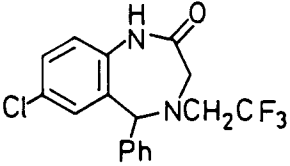
TABLE 6. N-Alkylation of Aromatic Amines

Entry	Substrate	Product	Conditions	Yield	Ref.
1		 R = H, Me, Et	NaBH_4 RCO_2H	72-83%	11
2	"	" R = Ph	NaBH_4 toluene PhCO_2H Δ	90%	12
3			NaBH_4 HOAc 50-60°	88%	11
4		 R = Me, Et	NaBH_4 RCO_2H THF	68-83%	53

SODIUM BOROHYDRIDE IN CARBOXYLIC ACID MEDIA. A REVIEW

5			NaBH ₄ HOAc 50-55°	85%	51
6	Ph ₂ NH	Ph ₂ NCH ₂ CH ₃	NaBH ₄ HOAc 60°	80%	11
7	"	Ph ₂ NCH ₂ CH ₂ Cl	NaBH ₄ ClCH ₂ CO ₂ H	90%	12
8			NaBH ₄ HOAc	92%	11
9			NaBH ₄ HOAc 60°	72%	11
10			NaBH ₄ HOAc rt, 7 hr	63%	54
11	"		NaBH ₄ HOAc rt, 5 days	71%	54
12			NaBH ₄ HOAc 20°	89%	51
13			NaBH ₄ HOAc 60°	81%	51
14			NaBH ₄ HOAc 15°	44%	55

15			NaBH_4 $\text{CH}_3(\text{CH}_2)_{14}\text{CO}_2\text{H}$	"low 56 yield"
16		 $\text{R} = \text{H}, \text{Me}$	NaBH_4 RCO_2H	70-95% 12
17		 $\text{R} = \text{H}, \text{Me}$	NaBH_4 RCO_2H	92% 57
18	 $\text{R} = \text{H}, \text{Me}, \text{Cl}$		NaBH_4 HCO_2H 0°	17-52% 58
19	 (others)		NaBH_4 HCO_2H	-- 58
20			NaBH_4 HCO_2H 10°	58% 28

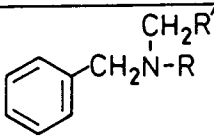
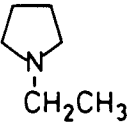
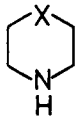
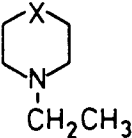
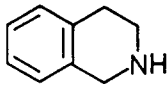
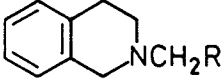
21			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$	7% 11,59
22			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$	61% 59
23			KBH_4 $\text{CF}_3\text{CO}_2\text{H}$	89% 47
24			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ toluene Δ	64% 13
25			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ 20%, 4 hr	25% 13

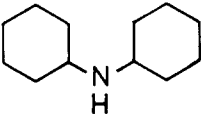
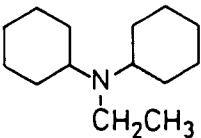
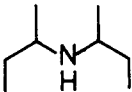
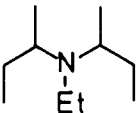
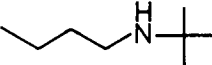
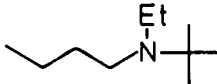
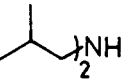
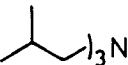
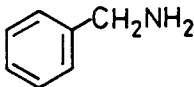
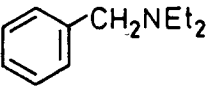
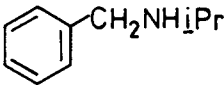
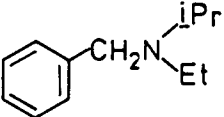
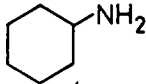
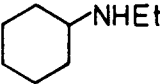
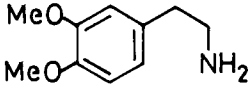
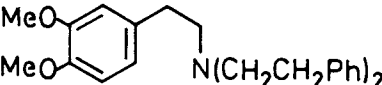
The N-alkylation of aliphatic amines using $\text{NaBH}_4/\text{RCO}_2\text{H}$ is tabulated in Table 7. It has proven to be a very general method with both primary and secondary amines and a variety of carboxylic acids (neat or in a cosolvent such as benzene). Hindered amines alkylate poorly (entries 8-10) or not at all (entry 28). The use of a ketone allows for the introduction of a secondary alkyl group (entry 13) or for the introduction of two different alkyl groups in converting a primary amine to a tertiary amine (entry 14). In some cases one can achieve N-monoalkylation of a primary amine (entries 15, 21).

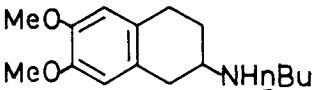
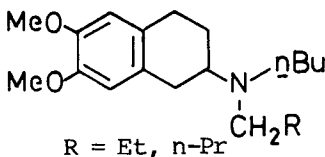
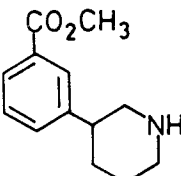
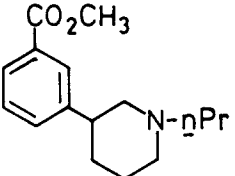
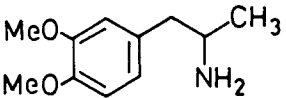
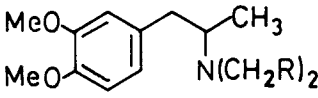
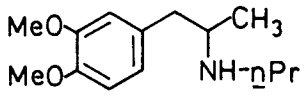
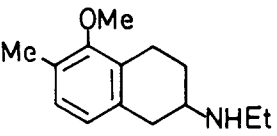
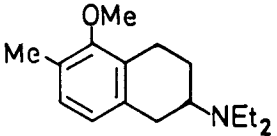
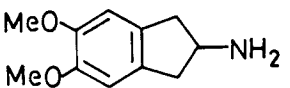
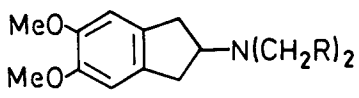
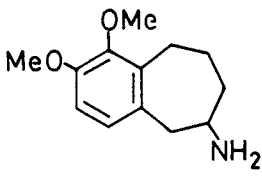
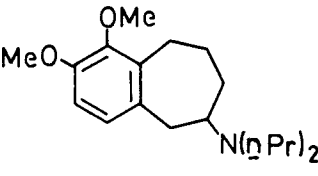
The Marchini modification¹² using benzene as a cosolvent has been widely used (entries 17-27) by three groups⁶²⁻⁷² to synthesize an array of dopamine analogues.

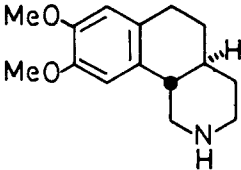
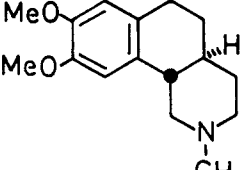
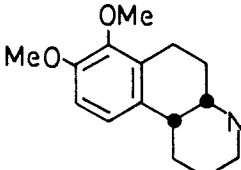
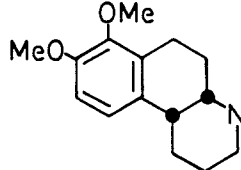
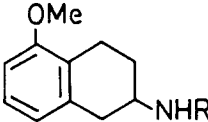
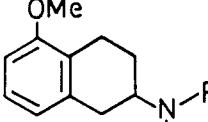
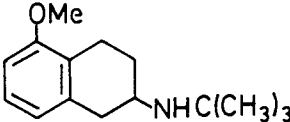
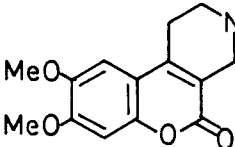
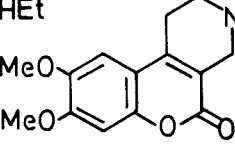
Fewer examples of N-methylation of aliphatic amines using $\text{NaBH}_4/\text{HCO}_2\text{H}$ have been reported (Table 6, entry 20), perhaps because alternative, well-established methods exist (e.g., $\text{HCHO}/\text{NaBH}_3\text{CN}$) and the reaction of NaBH_4 with neat formic acid is exceptionally vigorous and unpleasant to conduct.

TABLE 7. N-Alkylation of Aliphatic Amines

Entry	Substrate	Product	Conditions	Yield	Ref.
1	 $\text{R} = \text{Me, Et, CH}_2\text{Ph, i-Pr, t-Bu}$	 $\text{R}' = \text{Me, Et, n-Pr, i-Pr, t-Bu}$	NaBH_4 $\text{R}'\text{CO}_2\text{H}$ $50-55^\circ$	62-84%	60
2			NaBH_4 HOAc $50-55^\circ$	74%	60
3	 $\text{X} = \text{CH}_2, \text{O, NMe}$		NaBH_4 HOAc $50-55^\circ$	69-84%	60
4	$(\text{CH}_3)_2\text{NH}_2\text{Cl}$	$(\text{CH}_3)_2\text{N}(\text{CH}_2)_8\text{CH}_3$	NaBH_4 $\text{CH}_3(\text{CH}_2)_7\text{CO}_2\text{H}$ THF NaOAc $50-55^\circ$	78%	60
5	$(\text{CH}_3\text{CH}_2)_2\text{NH}$	$(\text{CH}_3\text{CH}_2)_2\text{N}(\text{CH}_2)_7\text{CH}_3$	NaBH_4 $\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{H}$ $50-55^\circ$	70%	60
6		 $\text{R} = \text{Me, Et}$	NaBH_4 RCO_2H	58-65%	53

7	$n\text{Bu}_2\text{NH}$	$n\text{Bu}_2\text{NEt}$	NaBH_4 HOAc 80° , 3 hr	80%	12
8			NaBH_4 HOAc $50-55^\circ$	14%	60
9			NaBH_4 HOAc $50-55^\circ$	13%	60
10			NaBH_4 HOAc $50-55^\circ$	9%	60
11			NaBH_4 $i\text{-Pr}-\text{CO}_2\text{H}$	90%	51
12			NaBH_4 HOAc $50-55^\circ$	66%	60
13	"		$\text{CH}_3\overset{\text{O}}{\parallel}\text{CCH}_3$ NaBH_4 HOAc 25°	84%	60
14	"		1. CH_3COCH_3 NaBH_4 HOAc , 25° 2. $50-55^\circ$	82%	60
15			NaBH_4 HOAc 45° , 4 hr	61%	51
16	$(\text{CH}_3)_2\text{NH}$	$(\text{CH}_3)_2\text{N}(\text{CH}_2)_{12}\text{N}(\text{CH}_3)_2$	NaBH_4 $\text{HO}_2\text{C}(\text{CH}_2)_{10}\text{CO}_2\text{H}$	21%	61
17			NaBH_4 $\text{PhCH}_2\text{CO}_2\text{H}$ PhH Δ 19 hr	51%	62

18		 R = Et, n-Pr	NaBH ₄ RCO ₂ H PhH Δ 3 hr	80-86%	62
19			NaBH ₄ EtCO ₂ H PhH Δ	72%	63
20		 R = Me, Et, n-Pr	(15 molar equiv.) NaBH ₄ RCO ₂ H PhH Δ, 16 hr	54-87%	64
21	"		(2 molar equiv.) NaBH ₄ EtCO ₂ H PhH Δ, 16 hr	38%	64
22			NaBH ₄ HOAc PhH Δ	96%	65
23		 R = Me, Et	NaBH ₄ RCO ₂ H PhH 20 hr, Δ	53-75%	66
24			NaBH ₄ EtCO ₂ H PhH CH ₂ Cl ₂	43%	67

25	 (<u>cis</u> also)	 R = Me, Et	NaBH ₄ RCO ₂ H PhH, Δ	83-86% 68
26	 (<u>trans</u> also)	 R = Me, Et	NaBH ₄ RCO ₂ H PhH Δ	89-91% 69
27	 R = Me, Et, n-C ₅ H ₁₁ , n-Pr, i-Pr	 R' = Me, Et, CH ₂ Ph	NaBH ₄ RCO ₂ H PhH Δ 5 hr	46-98% 70
28	 —	—	NaBH ₄ EtCO ₂ H	failed 70
29	 R = Me, Et, Pr, i-Bu	 R = Me, Et, Pr, i-Bu	NaBH ₄ RCO ₂ H	20-54% 132

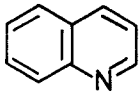
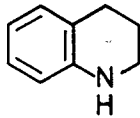
VII. REDUCTION AND REDUCTION/N-ALKYLATION OF π-DEFICIENT HETEROCYCLES

Following an early report by Rao and Jackman⁷³ on the reduction of nitroquinolines and related compounds with NaBH₄/ HOAc, numerous examples of the reduction of π-deficient

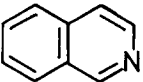
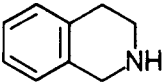
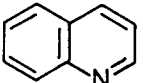
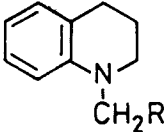
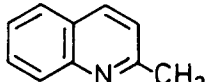
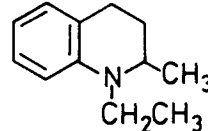
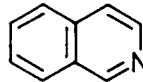
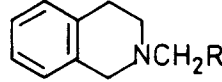
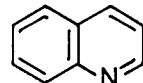
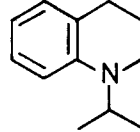
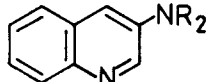
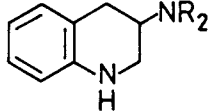
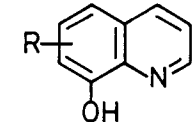
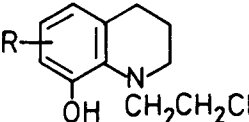
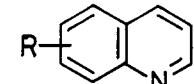
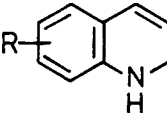
heterocycles using this methodology have been disclosed (Table 8). As has been seen earlier, the reaction can be controlled by changing either the borohydride reagent or the temperature to give reduction with or without N-alkylation (entries 1 and 3, 2 and 5, 9 and 10, 14 and 16, 19 and 20, 24 and 25). A ketone can be employed to give a secondary N-alkyl group (entry 6). Only in the case of nitroquinolines does the reduction stop at the 1,2-dihydroquinoline stage (entries 9, 10), although, in the presence of acetic anhydride, the 1,2-dihydro heterocycles can be trapped in the case of quinoline and isoquinoline (entries 12-13). The reduction of quinoline and isoquinoline with $\text{NaBH}_4/\text{CF}_3\text{CO}_2\text{H}$ (not shown in Table 8) gives a mixture of the corresponding 1,2,3,4-tetrahydro heterocycle and the N-trifluoroethylated derivative (17-21%).⁵³

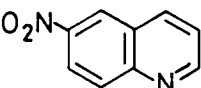
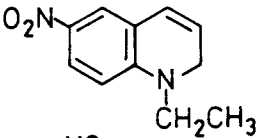
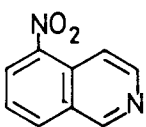
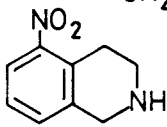
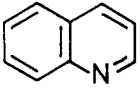
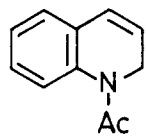
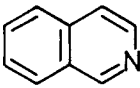
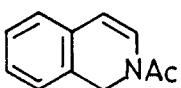
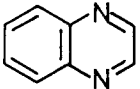
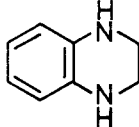
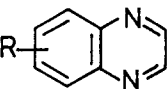
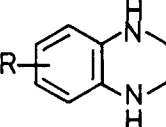
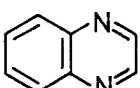
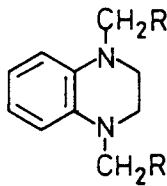
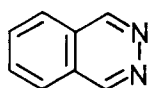
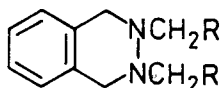
Although pyridine is not reduced with $\text{NaBH}_4/\text{RCO}_2\text{H}$, under conditions thus far investigated,⁵¹ pyridines containing 3,5-electron-withdrawing groups are smoothly reduced to the 1,4-dihydro compounds with $\text{NaBH}_3\text{CN}/\text{HOAc}$ (entries 27-29) but not with $\text{NaBH}_4/\text{HOAc}$ (entry 26).

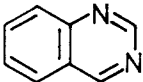
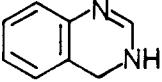
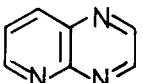
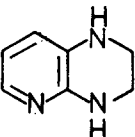
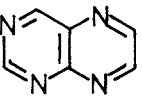
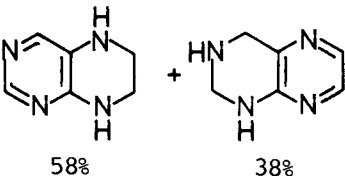
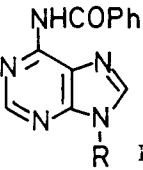
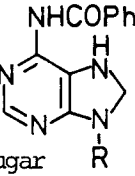
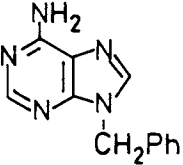
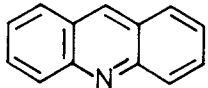
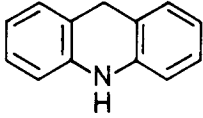
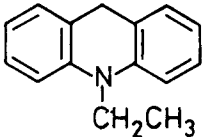
TABLE 8. Reduction and Reduction/N-Alkylation of π -Deficient Heterocycles

Entry	Substrate	Product	Conditions	Yield	Ref.
1			NaBH_3CN HOAc 50° , 1 hr	71%	53

SODIUM BOROHYDRIDE IN CARBOXYLIC ACID MEDIA. A REVIEW

2			NaBH_3CN HOAc 50° , 1 hr	71% 53
3		 $\text{R} = \text{H, Me, Et}$	NaBH_4 RCO_2H 50°	52-68% 53
4			NaBH_4 HOAc 50° , 2.5 hr	65% 53
5		 $\text{R} = \text{H, Me, Et}$	NaBH_4 RCO_2H 50°	76-79% 53
6			NaBH_4 HOAc CH_3COCH_3 50° , 1 hr	59% 53
7	 $\text{R} = \text{H, Me}$		NaBH_3CN HOAc 55° , 1.5 hr	44-66% 74
8	 $\text{R} = \text{H, 6-Me, 5-Cl}$ (2-Me failed)		NaBH_4 $\text{ClCH}_2\text{CO}_2\text{H}$ THF	45-68% 75
9	 $\text{R} = 5, 6, 7, 8\text{-NO}_2$		NaBH_4 HOAc 5°	67-90% 73

10			NaBH ₄ HOAc 50°	44%	53
11			NaBH ₄ HOAc 5°	65%	73
12			NaBH ₄ HOAc Ac ₂ O 60°, 2 hr	72%	76
13			NaBH ₄ HOAc Ac ₂ O 60°, 2 hr	80%	76
14			NaBH ₄ CF ₃ CO ₂ H THF rt, 1 hr	90%	77
15	 R = 5-NO ₂ , 6-NO ₂ , 6-CN, 6-CF ₃ , 6-CO ₂ Et		NaBH ₄ HOAc 5°	43-87%	73
16		 R = H, Me, Et	KBH ₄ RCO ₂ H Δ, 6 hr	70-87%	78
17		 R = H, Me, Et	KBH ₄ RCO ₂ H Δ, 6 hr	41-97%	78

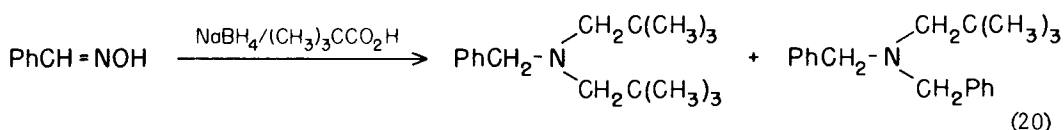
18			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ THF rt, 1 hr	85%	77
19			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ THF rt, 1 hr	75%	77
20	"	 R = H, Me, Et	KBH_4 RCO_2H Δ , 6 hr	43-66%	78
21		 58% 38%	NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ THF rt, 1 hr	96%	77
22	 R = CH_2Ph , sugar	 R	NaBH_4 HOAc rt, 20 min	58-75%	79
23		—	NaBH_4 HOAc rt, 20 min	failed	79
24			NaBH_4 HOAc 20°	96%	51
25	"		NaBH_4 HOAc 75°	82%	51

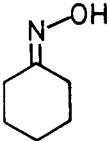
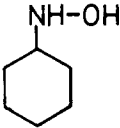
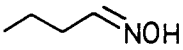
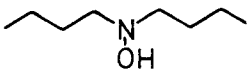
26	$\text{RO}_2\text{C}-\text{C}_5\text{H}_4-\text{CO}_2\text{R}$ $\text{RO}_2\text{C}-\text{C}_5\text{H}_4-\text{CO}_2\text{R}$ $\text{RO}_2\text{C}-\text{C}_5\text{H}_4-\text{CO}_2\text{R}$	NaBH_4 HOAc	57-76%	80
	$\text{R} = \text{Me}: 60\%$ $\text{R} = \text{Et}: 62\%$	40% 38%		
27	$\text{RO}_2\text{C}-\text{C}_5\text{H}_4-\text{CO}_2\text{R}$	NaBH_3CN HOAc	74-77%	80
	$\text{R} = \text{H}, \text{Me}, \text{Et}$			
28	$\text{NC}-\text{C}_5\text{H}_4-\text{CN}$ $\text{NC}-\text{C}_5\text{H}_4-\text{CN}$	NaBH_3CN HOAc	98%	80
29	$\text{EtO}_2\text{C}-\text{C}_5\text{H}_4-\text{CO}_2\text{Et}$ $\text{EtO}_2\text{C}-\text{C}_5\text{H}_4-\text{CO}_2\text{Et}$	NaBH_3CN HOAc	60%	80

VIII. REDUCTION AND REDUCTION/N-ALKYLATION OF OXIMES

Depending on the reaction conditions, oximes can be reduced either to N-monoalkylhydroxylamines or N,N-dialkylhydroxylamines, and oxime ethers can be reduced to primary amines (Table 9). The unsymmetrical dialkylhydroxylamines so prepared would be very difficult to synthesize other ways.

In some cases, aberrant reaction products are obtained, especially with aldoximes, where the initially-formed monoalkylhydroxylamine condenses with the oxime leading, after reduction, to the symmetrical dialkylhydroxylamine where both alkyl groups derive from the oxime (entry 6). Another side reaction is overreduction and subsequent alkylation, an example of which is shown in Eq. 20.⁸¹ Note that this particular reaction also gives a product of the type formed in entry 6.

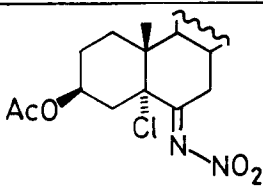
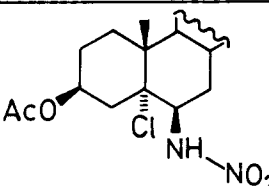
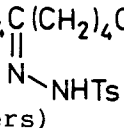
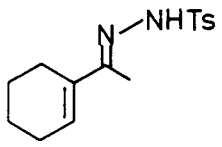
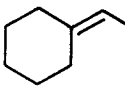
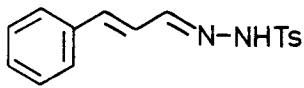
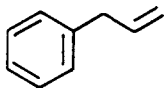
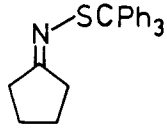
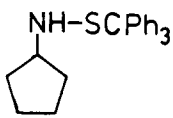
TABLE 9. Reduction and Reduction/N-Alkylation of Oximes, Oxime Ethers, and Oxime Esters

Entry	Substrate	Product	Conditions	Yield	Ref.
1	$\begin{array}{c} \text{N-OH} \\ \parallel \\ \text{R}-\text{C}-\text{R}' \end{array}$ R = H, Me R' = Me, Et, t-Bu, Ph, CH₂-Ph, n-Pr R,R' = -(CH₂)₅- R'' = Me, Et, i-Pr, n-Pr	$\begin{array}{c} \text{HO-NH-CH}_2\text{R}'' \\ \\ \text{R}-\text{CH}-\text{R}' \end{array}$	NaBH ₄ R''CO ₂ H 40-50° 4-5 hr	36-87%	81
2			NaBH ₃ CN HOAc 25° (NaBH ₄ HOAc 25°)	81% (63%)	81
3	$\begin{array}{c} \text{N-OCOPh} \\ \parallel \\ \text{R}-\text{C}-\text{R}' \end{array}$ R = CH₂Ph, R' = Me R,R' = -(CH₂)₄-, -(CH₂)₅-, -(CH₂)₁₁-	$\begin{array}{c} \text{NH-OCOPh} \\ \\ \text{R}-\text{CH}-\text{R}' \end{array}$	NaBH ₃ CN HOAc 20°	70-92%	82
4	$\begin{array}{c} \text{N-OCH}_3 \\ \parallel \\ \text{R}-\text{C}-\text{R}' \end{array}$ R = Ph, n-C₉H₁₉, CH₂CH₂Ph R' = Me, H, Ph R,R' = -(CH₂)₅-, -(CH₂)₆-	$\begin{array}{c} \text{NH}_2 \\ \\ \text{R}-\text{CH}-\text{R}' \end{array}$	NaBH ₄ CF ₃ CO ₂ H THF Δ, 2 hr	81-91%	83
5	$\begin{array}{c} \text{N-OH} \\ \parallel \\ \text{Ph}-\text{C}-\text{CH}_3 \end{array}$	$\begin{array}{c} \text{NH}_2 \\ \\ \text{Ph}-\text{CH}-\text{CH}_3 \end{array}$	NaBH ₄ CF ₃ CO ₂ H diglyme Δ, 5 hr	51%	83
6			NaBH ₄ HOAc 40-50° 2 hr	21%	81

IX. REDUCTION OF OTHER C=N COMPOUNDS

As might be anticipated from the results in the previous Section, a smattering of other C=N species have been reduced with $\text{NaBH}_4/\text{RCO}_2\text{H}$. These are tabulated in Table 10. Note-worthy is the convenient reductive deoxygenation of carbonyl compounds via their tosylhydrazones as developed by Hutchins and Natale⁸⁵ (entries 2-4).

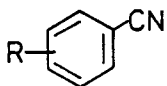
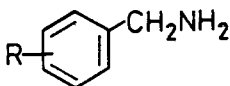
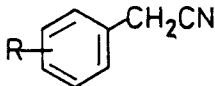
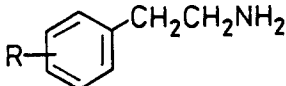
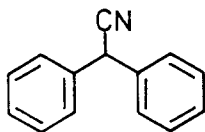
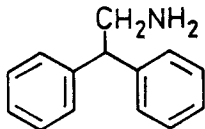
TABLE 10. Reduction of Other C=N Compounds

Entry	Substrate	Product	Conditions	Yield	Ref.
1	 (others)		NaBH_4 HOAc dioxane EtOH rt	76%	84
2	$\text{CH}_3(\text{CH}_2)_4\text{C}(\text{CH}_2)_4\text{CH}_3$  (others)	$\text{CH}_3(\text{CH}_2)_9\text{CH}_3$	NaBH_4 HOAc 70° 1-2 hr	84%	85
3	 (others)		NaBH_4 HOAc 70° 1-2 hr	72%	85
4			NaBH_4 HOAc 70° , 1-2 hr	56%	85
5	 (others)		NaBH_3CN $\text{CF}_3\text{CO}_2\text{H}$ THF	97%	86

X. REDUCTION OF NITRILES

Although nitriles are not reduced under conditions which produce $\text{NaBH}(\text{OCOR})_3$ (Table 8, entries 15 and 28; Table 15, entry 1), Umino and coworkers⁸⁷ have shown that nitriles are smoothly reduced to primary amines with $\text{NaBH}_3\text{OCOCF}_3$ (*in situ*) in THF at rt (Table 11). The reduction is poor with NaBH_3OAc .

Table 11. Reduction of Nitriles

Entry	Substrate	Product	Conditions	Yield	Ref.
1	 $\text{R} = \text{H}, 4\text{-Me}, 4\text{-CO}_2\text{Me}, 3\text{-NO}_2$		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ THF 20° , 4 hr	76-89%	87
2	 $\text{R} = \text{H}, 4\text{-NO}_2, 4\text{-Cl}$		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ THF 20° , 4 hr	70-71%	87
3			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ THF 20° , 4 hr	70%	87

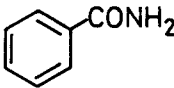
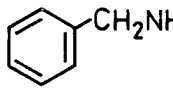
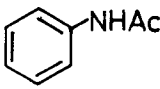
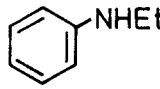
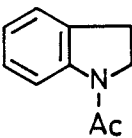
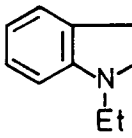
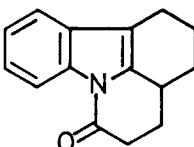
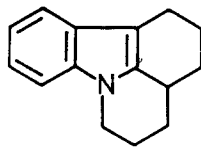
XI. REDUCTION OF AMIDES AND CARBAMATES

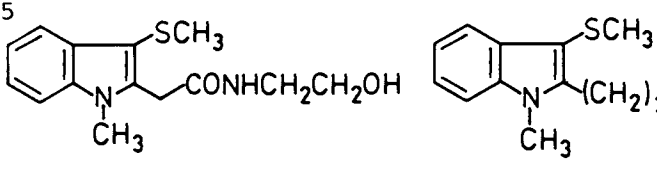
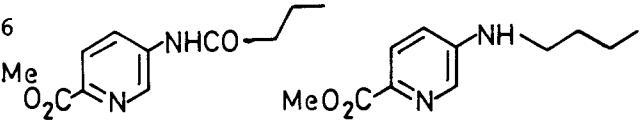
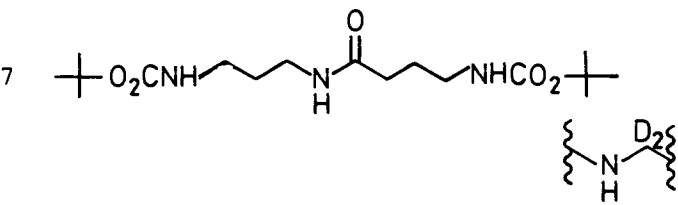
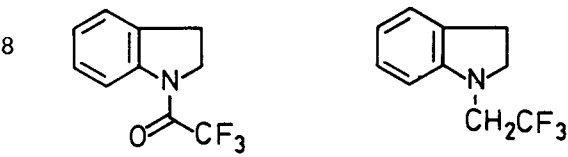
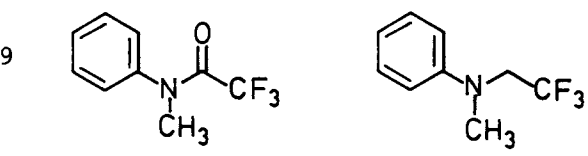
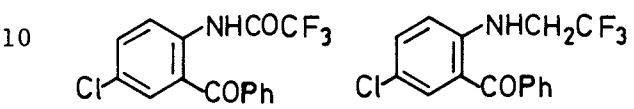
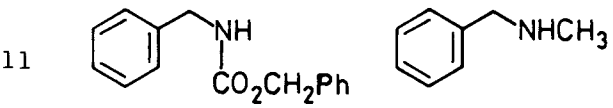
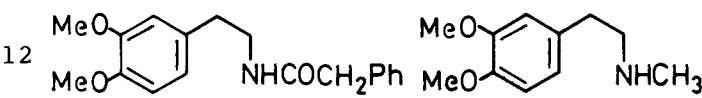
As is the case with nitriles (*vide supra*), amides are not reduced under conditions which produce $\text{NaBH}(\text{OCOR})_3$. For example, we determined that 1-acetylintole and 1-acetylintoline were not reduced to 1-ethylindoline to any appreciable extent under conditions which convert indole to 1-ethylindoline in high yield (NaBH_4 , excess HOAc).¹¹ For other examples of

amides (and similar carbonyls) that are not reduced under these conditions, see Table 3, entries 2, 3, 5; Table 4, entries 5, 7, 8, 14-16; Table 6, entry 25; Table 15, entries 1, 15.

However, Umino and coworkers⁸⁸ have shown that the more reactive NaBH_3OCOR ($\text{R} = \text{CH}_3, \text{CF}_3$) are capable of reducing amides and carbamates to amines (Table 12). Tertiary amides require $\text{NaBH}_3\text{OCOCF}_3$ for reduction (entries 3, 4), whereas primary and secondary amides are reduced by NaBH_3OAc . Although carbamates can also be reduced under these conditions (entries 11, 12), the t-BOC protecting group survived intact in the reduction of an amide with $\text{NaBD}_3\text{OCOCF}_3$ (entry 7).

TABLE 12. Reduction of Amides

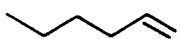
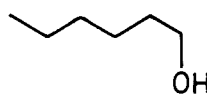
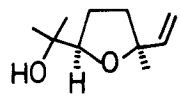
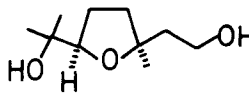
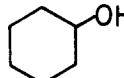
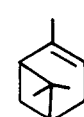
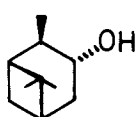
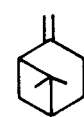
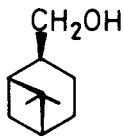
Entry	Substrate	Product	Conditions	Yield	Ref.
1			NaBH_4 $\text{CH}_3\text{CO}_2\text{H}$ dioxane Δ , 4 hr	76%	88
2			NaBH_4 $\text{CH}_3\text{CO}_2\text{H}$ dioxane Δ , 1 hr	88%	88
3			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ dioxane Δ , 5 hr	64% (28% with HOAc)	88
4			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ THF Δ , 4 hr	83%	89

5		NaBH_4 HOAc dioxane Δ , 2 hr	80%	90
6		NaBH_4 HOAc dioxane Δ , 30 min	69%	91
7		$\text{NaBD}_3\text{O}_2\text{CCF}_3$	--	92
8		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ THF	58%	59
9		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ THF	32%	59
10		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ tol, 6 hr 30-40°	76%	13
11		NaBH_4 HOAc dioxane Δ , 5 hr	66%	88
12		NaBH_4 HOAc dioxane Δ , 5 hr	82%	88

XII. HYDROBORATION OF ALKENES

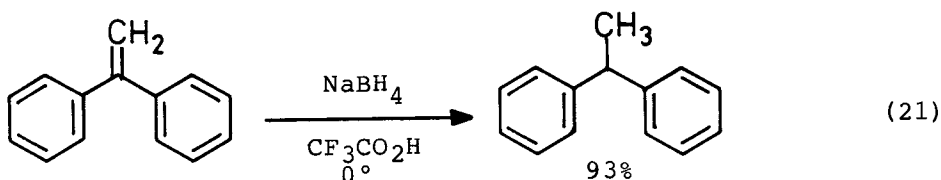
The second reported use of $\text{NaBH}_4/\text{RCO}_2\text{H}$ in synthesis--also described by Marshall and Johnson⁹³--was the hydroboration of alkenes. Although this method has not been widely used as such, several examples are known (Table 13). It is presumed that the hydroborating agent is NaBH_3OAc from the work of Hach⁹⁵ who optimized the reaction conditions. This would explain the apparent lack of hydroboration of alkenes under conditions that generate $\text{NaBH}(\text{OCOR})_3$ (e.g., Table 6, entry 5; Table 10, entries 3, 4).

TABLE 13. Hydroboration of Alkenes

Entry	Substrate	Product	Conditions	Yield	Ref.
1			1. NaBH_4 HOAc THF 2. H_2O_2 , OH^-	75%	93
2			1. NaBH_4 HOAc THF 2. H_2O_2 , OH^-	79%	94
3	 (others)		1. NaBH_4 HOAc THF 10-20° 2. H_2O_2 , OH^-	85%	95
4	 (others)		1. NaBH_4 THF HOAc, 20° 2. H_2O_2 , OH^-	73%	96a
5			1. LiBH_4 THF, 20° HOAc 2. H_2O_2 , OH^-	95%	96a

XIII. REDUCTION OF ALKENES

A second reaction of alkenes with $\text{NaBH}_4/\text{RCO}_2\text{H}$ that has been observed in one case is reduction (Eq. 21).⁹⁷ Thus far, this alkene reduction is restricted to alkenes that can form a resonance-stabilized carbocation (e.g., doubly benzylic) in trifluoroacetic acid (TFA).



The use of TFA in this regard is discussed further in the next two sections.

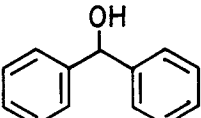
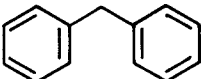
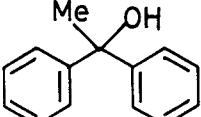
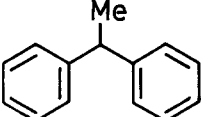
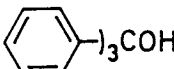
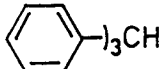
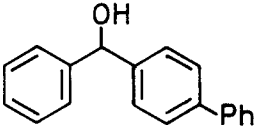
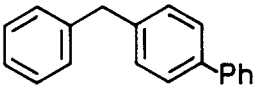
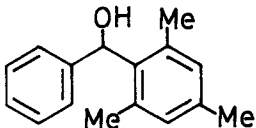
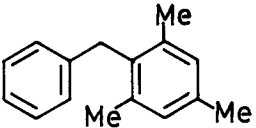
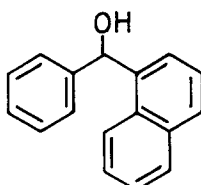
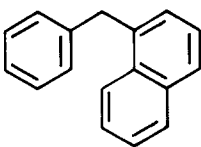
XIV. REDUCTION OF ALCOHOLS

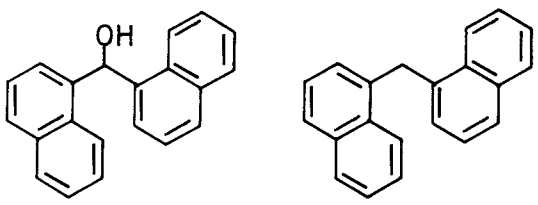
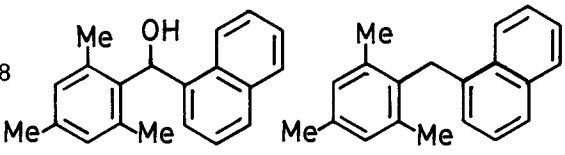
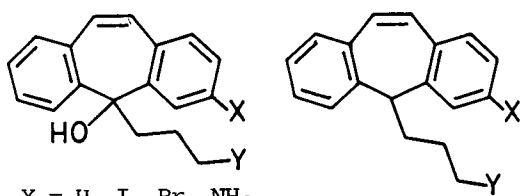
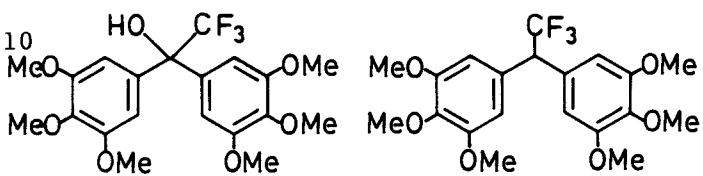
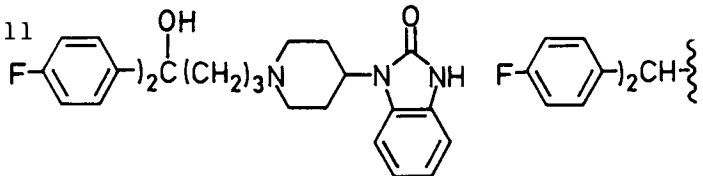
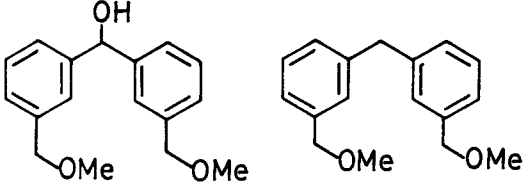
Trifluoroacetic acid, which is an excellent solvent for solvolysis and other $\text{S}_{\text{N}}1$ reactions (ionizing power Y value = 1.84⁹⁸), proves to be an ideal solvent with which to reduce diarylmethanols and triarylmethanols to the corresponding hydrocarbons with NaBH_4 .⁹⁷ This reduction method has proven to be exceedingly general and highly efficient (Table 14). Although yields are generally lower for monobenzylic alcohols (entries 20-22), in some cases it has been very successful (entries 17, 18). Reduction of benzyl alcohol, 1- and 2-octanol, and 1-methylcyclohexanol under these conditions is not observed.⁹⁷ The reduction is very slow or fails in glacial HOAc , at least with triphenylmethanol.⁹⁷

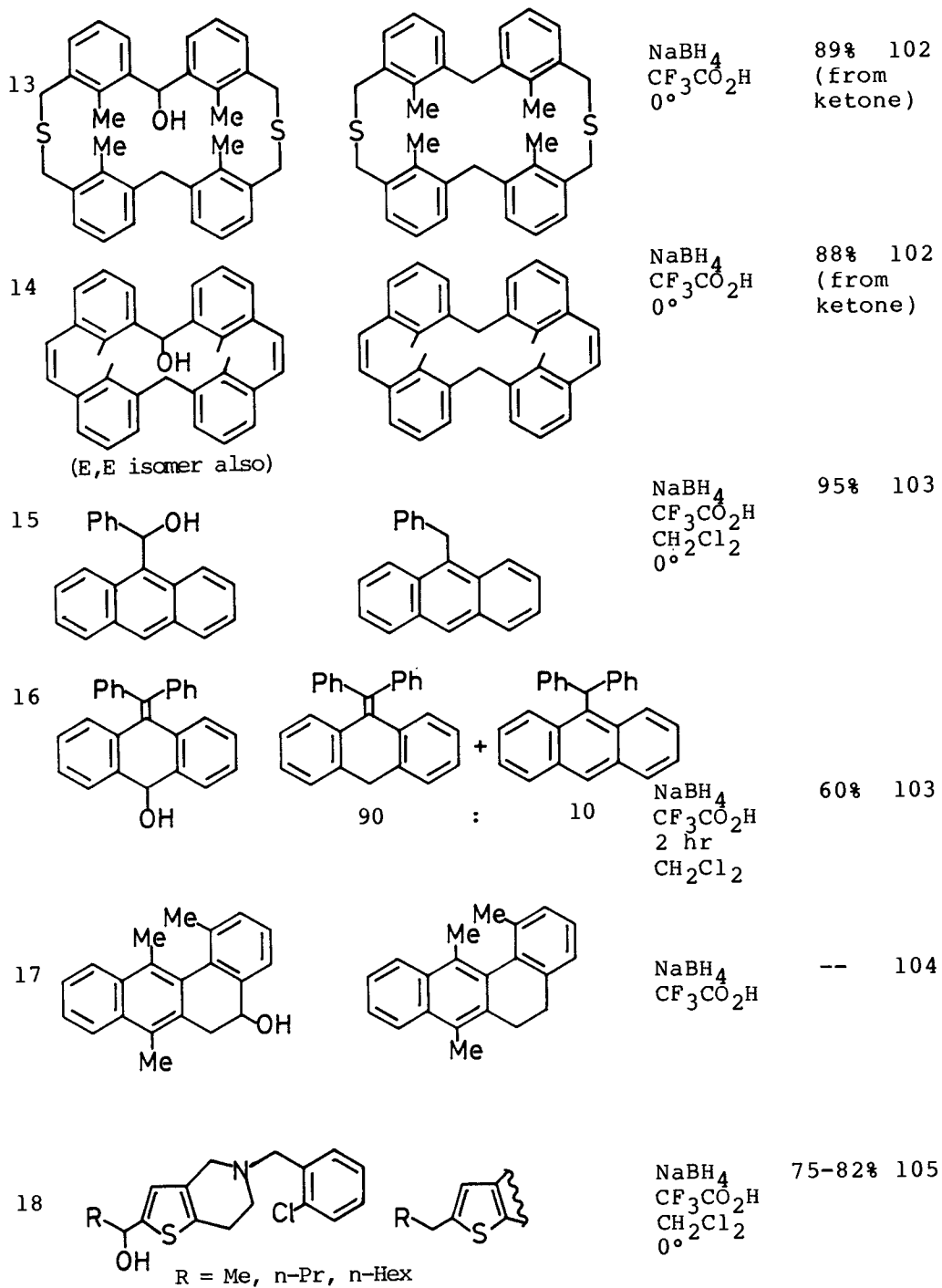
In most of the cases that we have studied,⁹⁷ the reaction is complete in seconds and can be monitored visually. Thus, the carbocation, which is usually highly colored, forms instantly as the alcohol is added to the suspension of NaBH_4

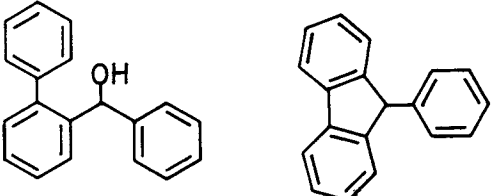
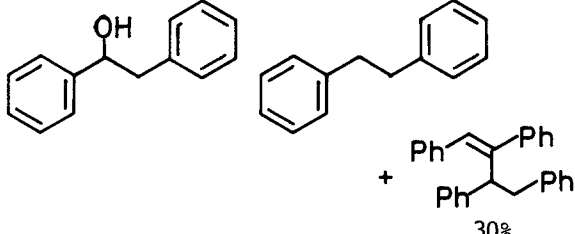
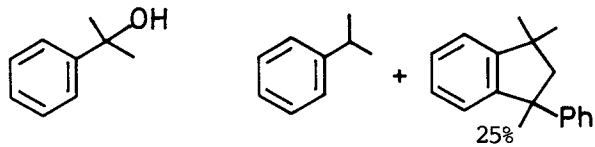
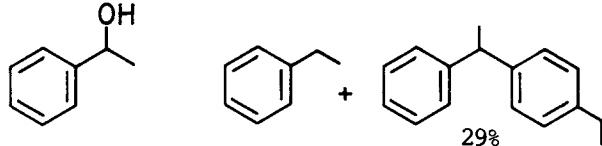
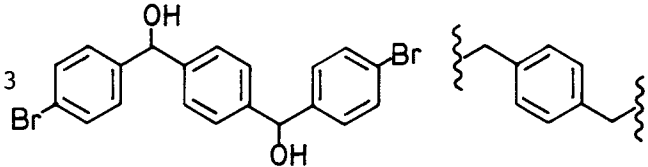
in TFA, but then is rapidly quenched (color disappears) to give product. In one case (entry 19), the intermediate carbocation cyclizes faster than it undergoes reduction. In the case of several monobenzylic alcohols (entries 20-22), other products, resulting from dehydration and dimerization (entries 20, 21) or alkylation of the product by the carbocation (entry 22), are observed.

TABLE 14. Reduction of Alcohols

Entry	Substrate	Product	Conditions	Yield	Ref.
1			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ 15-20°	93%	97
2			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ 0° 5 min	97%	97
3			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ CH_2Cl_2 15-20°	99%	97
4			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ 15-20° CH_2Cl_2	94%	97
5			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ CH_2Cl_2 15-20°	90%	97
6			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ CH_2Cl_2 15-20°	86%	97

- 7  NaBH_4 94% 97
 $\text{CF}_3\text{CO}_2\text{H}$
 CH_2Cl_2
 $15-20^\circ$
- 8  NaBH_4 77% 97
 $\text{CF}_3\text{CO}_2\text{H}$
 CH_2Cl_2
 $15-20^\circ$
- 9  NaBH_4 98-100% 97
 $\text{CF}_3\text{CO}_2\text{H}$
 0°
5 min
X = H, I, Br, NH_2
Y = NMe_2 , OEt
- 10  NaBH_4 78% 99
 $\text{CF}_3\text{CO}_2\text{H}$
- 11  NaBH_4 -- 100
 $\text{CF}_3\text{CO}_2\text{H}$
 CH_2Cl_2
2 hr
- 12  NaBH_4 94% 101
 $\text{CF}_3\text{CO}_2\text{H}$
 0°



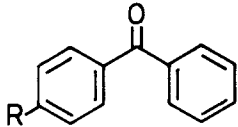
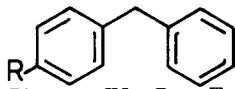
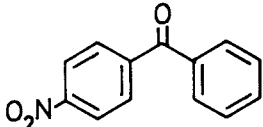
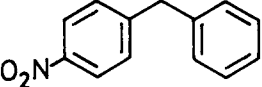
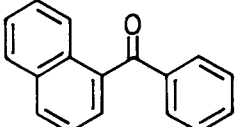
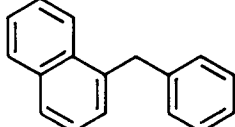
19		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ CH_2Cl_2 15-20°	93%	97
20		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ 0°	70%	97
21		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ 15-20°	45%	97
22		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ 15-20°	9%	97
23		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ CH_2Cl_2 15-20°	97%	51

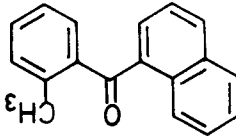
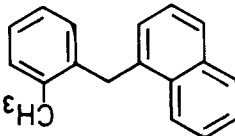
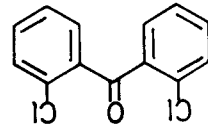
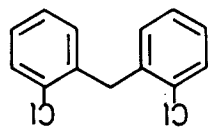
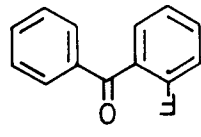
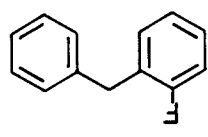
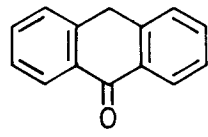
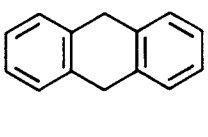
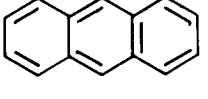
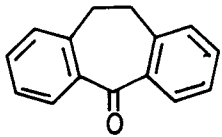
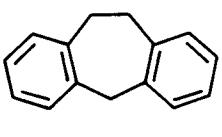
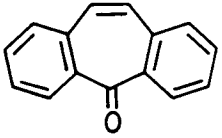
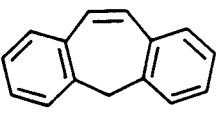
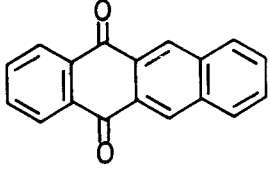
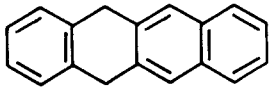
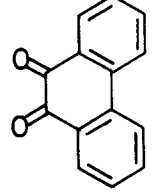
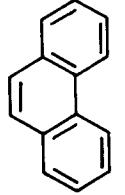
XV. REDUCTION OF KETONES TO HYDROCARBONS

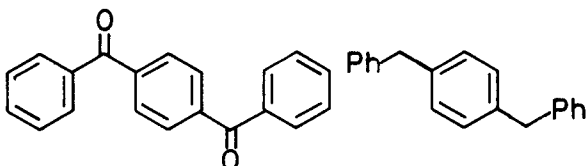
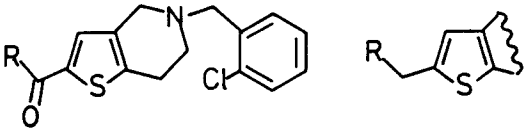
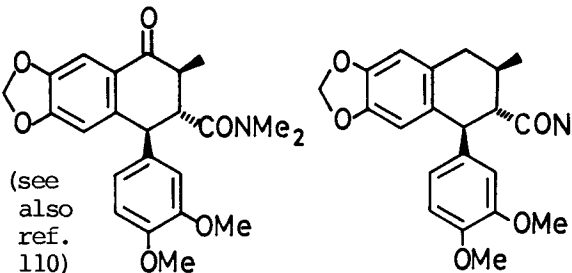
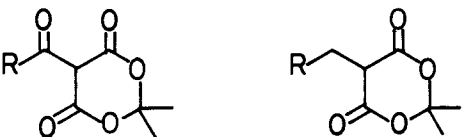
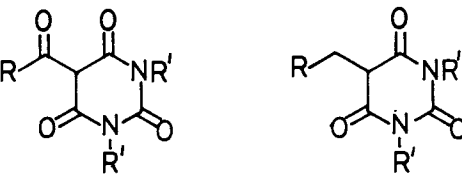
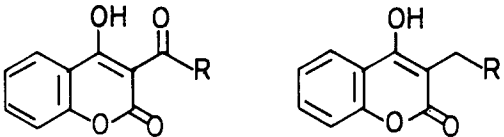
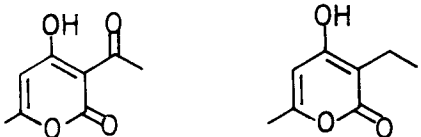
The companion reaction to the reduction of diarylmethanols to diarylmethanes with NaBH_4/TFA (Section XIV) is the reduction of diarylketones to diarylmethanes under the same conditions (Table 15). This reaction is very efficient and general, and in some cases works well for monoaryl ketones (entries 14, 15, 22). However, Michler's ketone (4,4'-bis-[dimethylamino]benzophenone) and decafluorobenzophenone fail to

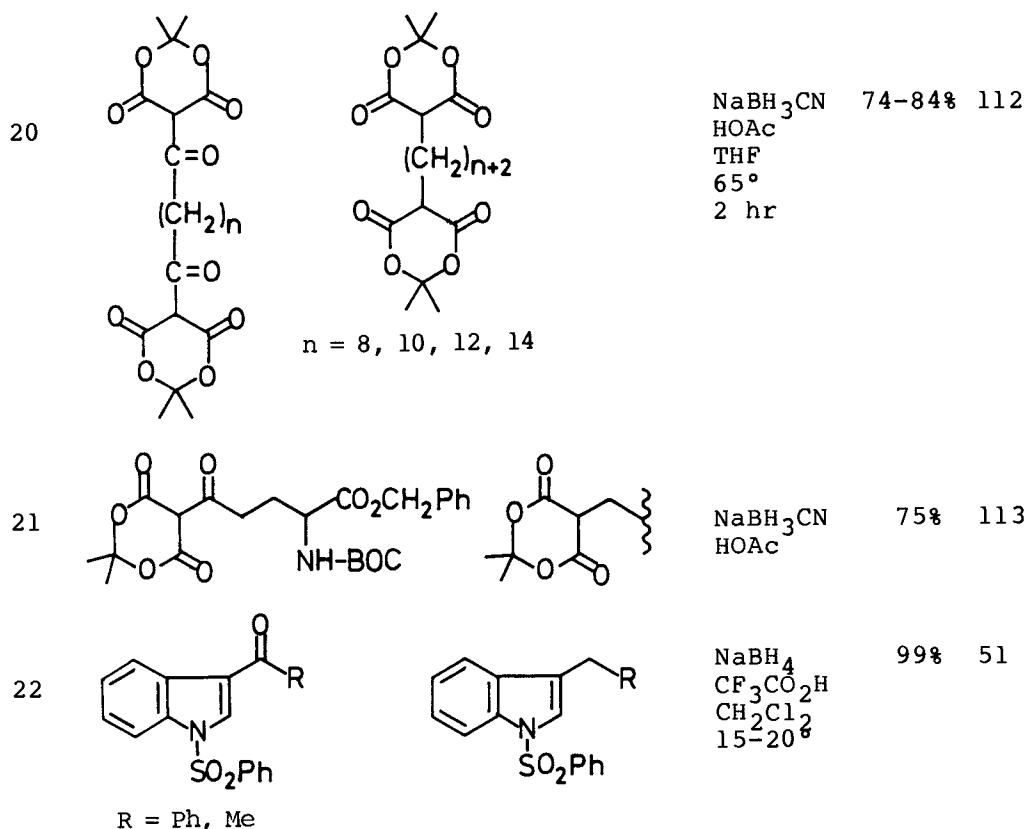
react with $\text{NaBH}_4/\text{CF}_3\text{CO}_2\text{H}$, and the reduction of 4-nitrobenzophenone (entry 2) is very slow. Likewise, the sterically hindered diarylketones mesityl phenyl ketone, dimesityl ketone, and mesityl α -naphthyl ketone give little or no reduction product.¹⁰⁶ Depending on the mode of addition, anthrone may be reduced either to dihydroanthracene (entry 7) or to anthracene (entry 8). In unpublished work, we have found that quinones are reduced either to a fully reduced compound (entry 11) or to the corresponding aromatic hydrocarbon (entry 12). 1,4-Naphthoquinone and 9,10-anthraquinone are also reduced to their respective aromatic hydrocarbons in variable yields.⁵¹ Smith and coworkers¹¹¹⁻¹¹² have developed a facile two-carbon homologation sequence using the $\text{NaBH}_3\text{CN}/\text{HOAc}$ reduction of acylated Meldrum's acid and related derivatives (entries 16-21).

TABLE 15. Reduction of Ketones to Hydrocarbons

Entry	Substrate	Product	Conditions	Yield	Ref.
1	 $\text{R} = \text{H, Me, OH, CO}_2\text{H, CO}_2\text{Me, CN, Br, F, NHCOPh, NMe}_2, \text{OMe}$	 $\text{R} = \text{Me, CO}_2\text{Me, CN, Br, F}$	NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ CH_2Cl_2 $15-20^\circ$	73-94%	106
2		 (+ 53% alcohol)	NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ CH_2Cl_2 $15-20^\circ$	43%	106
3			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ CH_2Cl_2 $15-20^\circ$	91%	106

4			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ CH_2Cl_2 15-20%	91%	106
5			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$	--	107
6			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ CH_2Cl_2 15-20%	85%	106
7			NaBH_4 (added last) $\text{CF}_3\text{CO}_2\text{H}$ 15-20%	91%	106
8	"		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ CH_2Cl_2 15-20%	78%	106
9			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ CH_2Cl_2 15-20%	88%	106
10			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$	85%	108
11			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ 15-25%	57%	51
12			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ 10-20%	23%	51

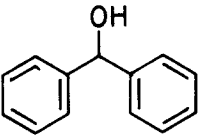
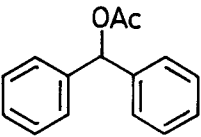
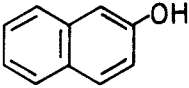
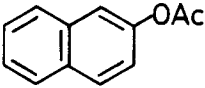
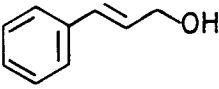
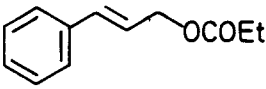
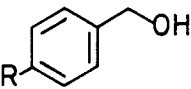
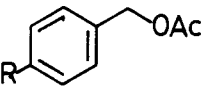
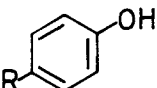
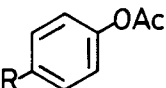
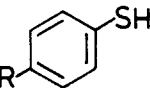
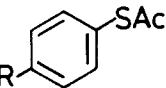
- 13  NaBH_4 86% 51
 $\text{CF}_3\text{CO}_2\text{H}$
 $15-20^\circ$
- 14  NaBH_4 75-88% 105
 $\text{CF}_3\text{CO}_2\text{H}$
 CH_2Cl_2
 0°
- R = Me, n-Pr, n-Hex
- 15  NaBH_4 76% 109
 $\text{CF}_3\text{CO}_2\text{H}$
 $0^\circ \rightarrow 25^\circ$
 (see also ref. 110)
- 16  NaBH_3CN 50-85% 111
 HOAc
- R = Et, i-Pr, CH_2Ph
- 17  NaBH_3CN 75-96% 111
 HOAc
- R = n-Pr, i-Bu, n-Pen, $\text{n-C}_{15}\text{H}_{31}$
 R' = H, Me
- 18  NaBH_3CN 78-86% 111
 HOAc
- R = Et, n-Pr, n-Bu, $\text{n-C}_{15}\text{H}_{31}$
- 19  NaBH_3CN 70% 111
 HOAc



XVI. ACYLATION OF ALCOHOLS AND AMINES

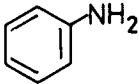
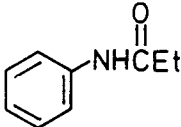
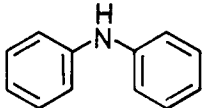
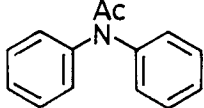
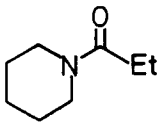
In what could be considered as a side-reaction in the chemistry of $\text{NaBH}_4/\text{RCO}_2\text{H}$, the acylation of suitable functional groups (e.g., alcohols, phenols, amines) is frequently encountered. Indeed, the isolation of methyl formate by Wartik and Pearson⁶ (Section I) is an example of the acylation (formylation) of methanol by a formyloxyboron species. Apparently independently, two groups have developed this into a useful alcohol and phenol acylation method (Table 16). It is presumed that under the reaction conditions (excess RCO_2H , reflux, 3 hr)¹² the acylating agent is NaB(OCOR)_4 or even B(OCOR)_3 (plus NaO_2CR).

TABLE 16. Acylation of Alcohols, Phenols and Thiophenols

Entry	Substrate	Product	Conditions	Yield	Ref.
1			NaBH_4 HOAc Δ , 3 hr	95%	12
2			NaBH_4 HOAc Δ , 3 hr	50%	12
3			NaBH_4 EtCO_2H $85-90^\circ$ 3 hr	80%	114
4	 $\text{R} = \text{H}, \text{OMe}, \text{NO}_2$		NaBH_4 HOAc $85-90^\circ$ 3 hr	90-95%	114
5	 $\text{R} = \text{H}, \text{Me}, \text{Cl}$		NaBH_4 HOAc Δ , 12 hr	80-98%	114
6	 $\text{R} = \text{H}, \text{Me}$		NaBH_4 HOAc Δ , 12 hr	75-95%	114

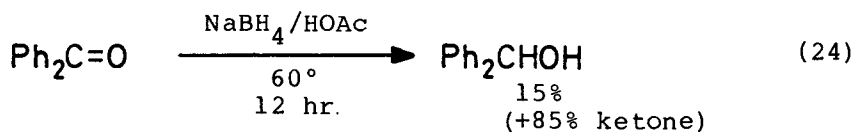
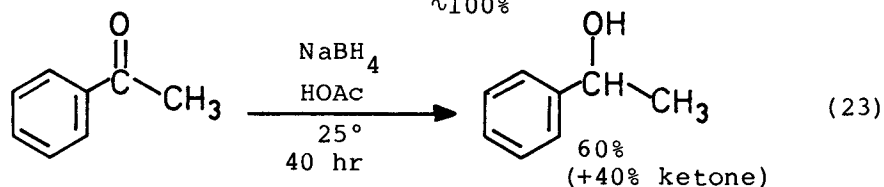
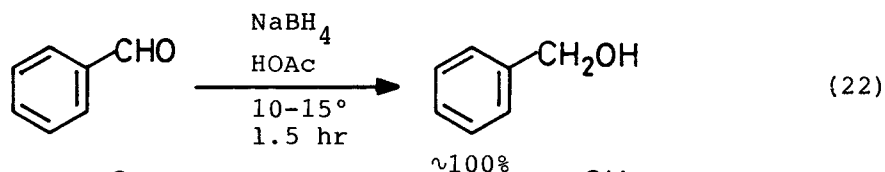
Amines can be similarly acylated to form amides (Table 17).

TABLE 17. Acylation of Amines

Entry	Substrate	Product	Conditions	Yield	Ref.
1			NaBH_4 EtCO_2H Δ , 3 hr	95%	12
2			NaBH_4 HOAc Δ , 3 hr	40%	12
3			NaBH_4 EtCO_2H Δ , 3 hr	60%	12

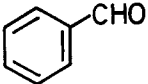
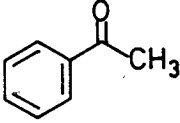
XVII. REDUCTION OF ALDEHYDES AND KETONES TO ALCOHOLS

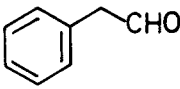
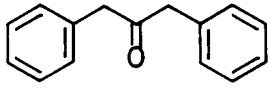
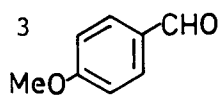
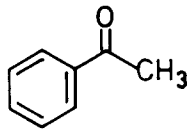
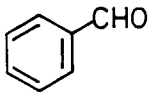
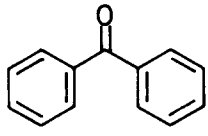
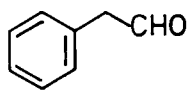
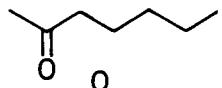
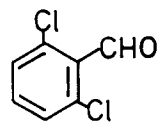
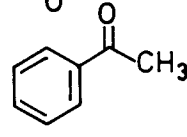
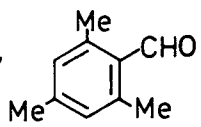
Early in our exploration of the chemistry of $\text{NaBH}_4/\text{RCO}_2\text{H}$, we observed that aldehydes and, especially, ketones are reduced more slowly to alcohols by NaBH_4 in glacial acetic acid than in alcoholic solution. For example, although benzaldehyde is completely reduced to benzyl alcohol, acetophenone and benzophenone are incompletely reduced to their alcohols with a large excess of NaBH_4 in glacial acetic acid (Eqs. 22-24).⁵¹ Even after these long reaction periods active borohydride reagent is present at the end of the reaction. In contrast, both of these ketones are rapidly and completely reduced to their respective alcohols with NaBH_4 in ethanol.



These observations paved the way for the chemoselective reduction of aldehydes, in the presence of ketones, using $\text{NaBH}(\text{OAc})_3$ in benzene¹¹⁵ or, even better, $n\text{-Bu}_4\text{NBH}(\text{OAc})_3$ in benzene.¹¹⁶ In both cases excess hydride reagent can be used. Examples of this chemoselective reduction of aldehydes to primary alcohols, in the presence of ketones, are tabulated in Table 18.

TABLE 18. Reduction of a 1:1 Mixture of Aldehyde and Ketone With $n\text{Bu}_4\text{NBH}(\text{OAc})_3$ in Benzene (24 hr, reflux)¹¹⁶

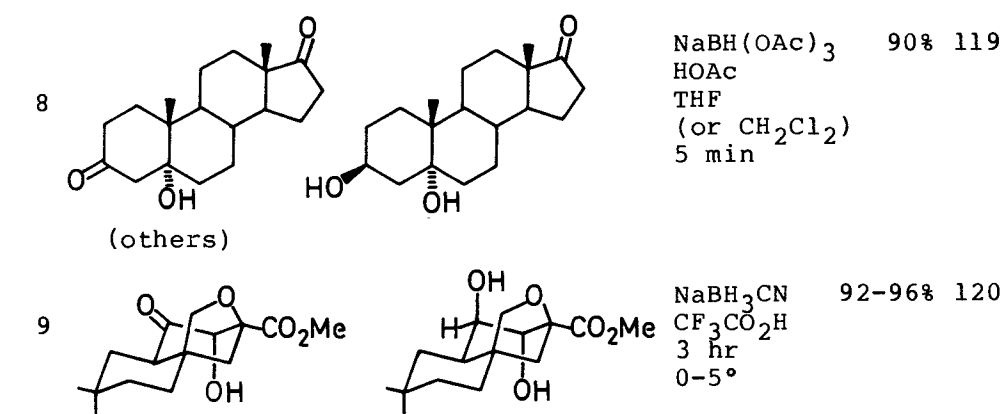
Entry	Aldehyde	Ketone	Yield of Primary Alcohol	Yield of Recovered Ketone
1			95%	96%

2			96%	94%
3			95%	99%
4			90%	94%
5			92%	96%
6			80%	92%
7		"	87%	96%

Moreover, as shown in Table 19, several ketoaldehydes have been reduced selectively to ketoalcohols or, in those cases where the hydroxyl group can complex with the borohydride species, to 1,3-diols (entry 4). Indeed, this method has been used by Saksena¹¹⁹ to reduce β -hydroxyketones to 1,3-diols with complete stereoselectivity (OH-assisted hydride delivery) (entries 7, 8). A related reduction has been described by Fuchs,¹²⁰ involving an α -hydroxyketone (entry 9).

TABLE 19. Reduction of Ketoaldehydes and Related Systems

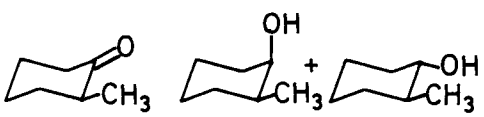
Entry	Substrate	Product	Conditions	Yield	Ref.
1			$\frac{n\text{-Bu}_4\text{NBH}(\text{OAc})_3}{\text{PhH}}$ Δ , 24 hr	88%	116
2			$\frac{n\text{-Bu}_4\text{NBH}(\text{OAc})_3}{\text{PhH}}$ Δ , 24 hr	72%	116
3			$\frac{n\text{-Bu}_4\text{NBH}(\text{OAc})_3}{\text{PhH}}$ Δ , 24 hr	77%	116
4			$\frac{n\text{-Bu}_4\text{NBH}(\text{OAc})_3}{\text{PhH}}$ Δ , 24 hr	80%	116
5			$\text{KBH}(\text{OAc})_3$ PhH	60%	117
6			NaBH_3CN HCO_2H t-BuOH	--	118
7			$\text{NaBH}(\text{OAc})_3$ HOAc rt	96%	119



Several groups have examined the stereochemistry of cyclic ketone reduction using $\text{NaBH}_4/\text{RCO}_2\text{H}$ (Table 20). Although the reduction of cyclohexanones is only moderately stereoselective with $\text{NaBH}_4/\text{HOAc}$, generally favoring the equatorial alcohol (entries 2, 3, 6), the stereoselectivity can be greatly enhanced by using acyloxyborohydride reagents derived from mandelic acid (entries 1, 5) or tartaric acid (entries 4, 7).

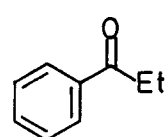
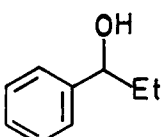
TABLE 20. Reduction of Cyclic Ketones

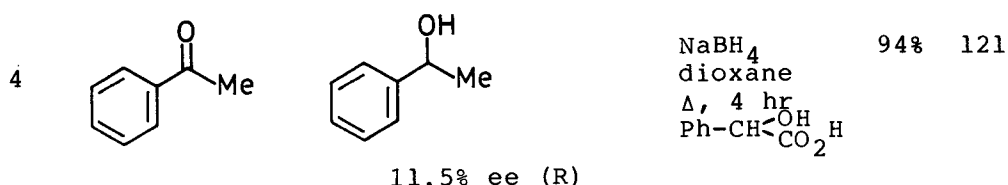
Entry	Substrate	Product	Conditions	Yield	Ref.
1	 (others)	 8 : 92	NaBH_4 $\text{PhCH(OH)CO}_2\text{H}$ <i>i</i> -PrOH, Δ 2 hr	--	121
2	"	26 : 74	NaBH_4 HOAc	90%	19
3	"	23 : 77	NaBH_3CN HOAc	95%	19

4	"	6 : 94	NaBH ₄ tartaric acid THF	122
5		25 : 75	NaBH ₄ PhCH(OH) CO ₂ H i-PrOH, Δ, 2 hr	121
6	"	55 : 45	NaBH ₄ HOAc	19
7	"	20 : 80	NaBH ₄ tartaric acid THF	122

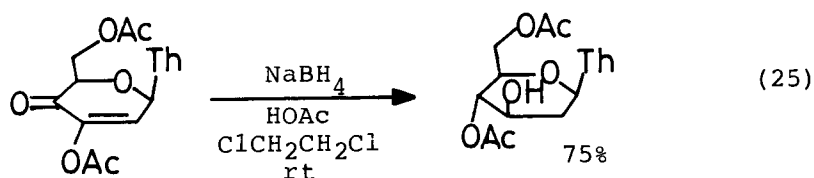
Several groups have examined the asymmetric reduction of ketones with optically active acyloxyborohydrides (Table 21), in some cases achieving good enantioselectivity. For each study, only the best of several systems examined is shown in Table 21.

TABLE 21. Asymmetric Reduction of Ketones

Entry	Substrate	Product	Conditions	Yield	Ref.
1	 (others)	 63% ee (R)	NaBH ₄ i-PrCO ₂ H THF, 25° sugar	56%	123
2	"	" 51% ee (R)	NaBH ₄ PhCH(OH) Et, THF 2 hr, rt sugar	68%	124
3	"	" 50% ee (S)	NaBH ₄ THF, rt 10 days proline	92%	125



Finally, the interesting double reduction of the ketone and enol acetate functionalities in a nucleoside has been reported, accompanied by an acetyl transposition (Eq. 25).¹²⁶

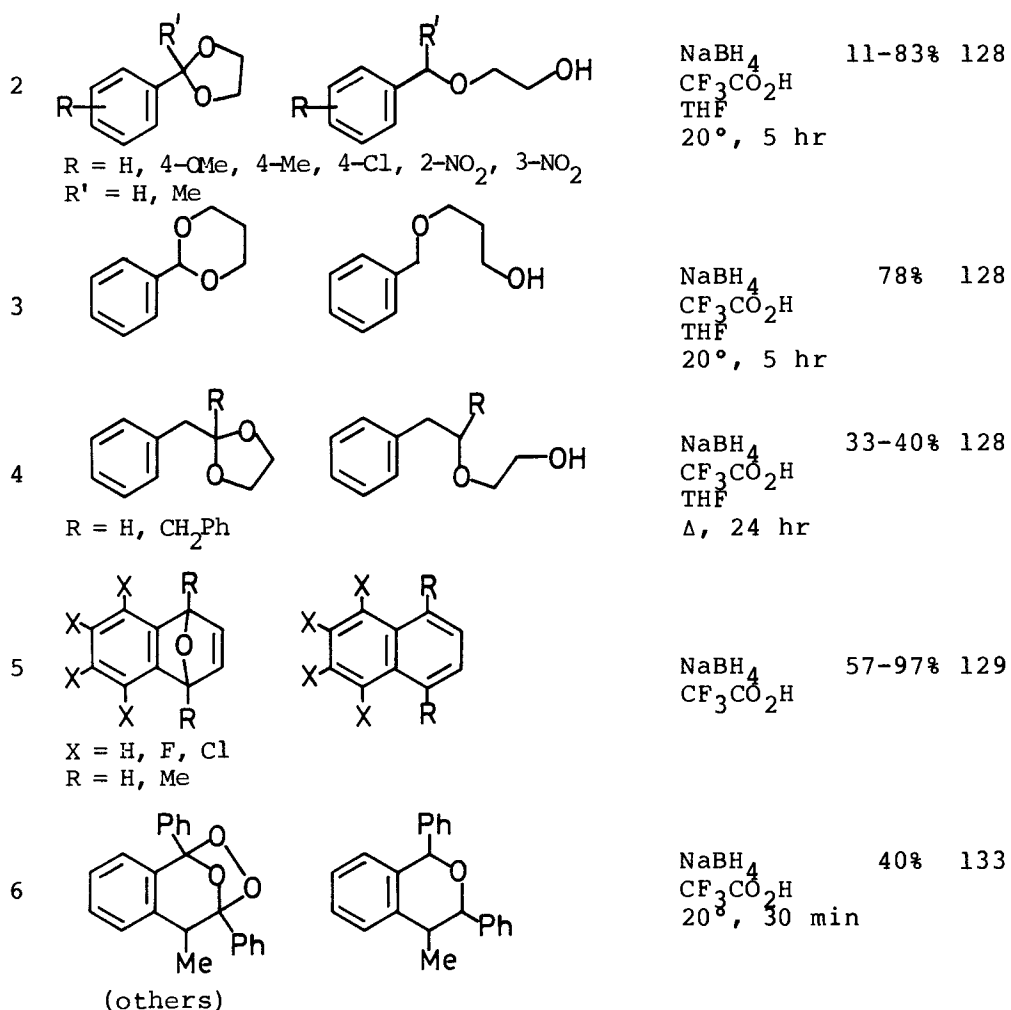


XVIII. REDUCTIVE CLEAVAGE OF ACETALS, KETALS, AND ETHERS

As might be expected, the use of trifluoroacetic acid in combination with NaBH₄ can effect the reductive cleavage of acetals, ketals, and ethers. A few examples have been reported (Table 22). The yields are higher for those systems giving rise to phenyl-stabilized oxonium ions (entries 1-3, 5 vs. entry 4). Recently, the deoxygenation of an ozonide was reported (entry 6).¹³³

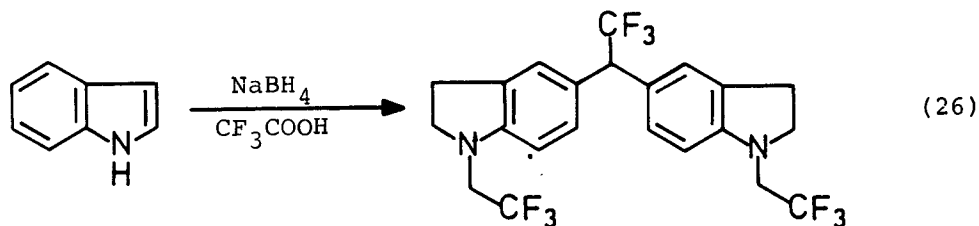
TABLE 22. Reductive Cleavage of Acetals, Ketals, and Ethers

Entry	Substrate	Product	Conditions	Yield	Ref.
1	 (others) Ph OMe Ph	 Ar HO Ph OMe Ph	NaBH ₃ CN CF ₃ CO ₂ H DMF	85%	127



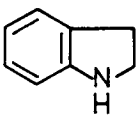
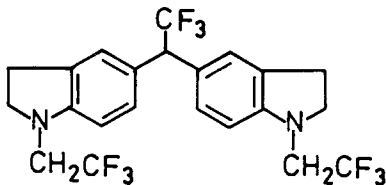
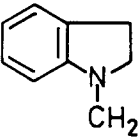
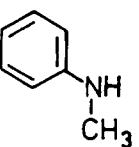
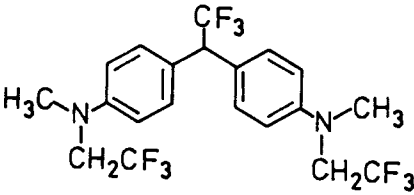
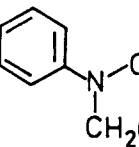
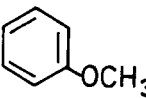
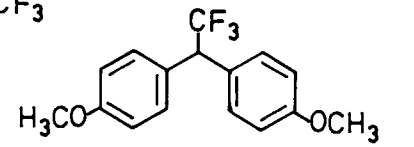
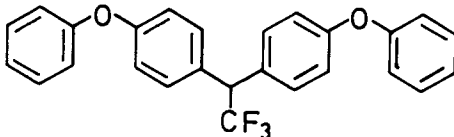
XIX. FRIEDEL-CRAFTS ALKYLATION OF ARENES

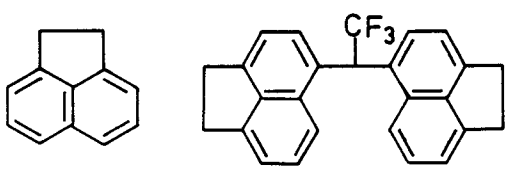
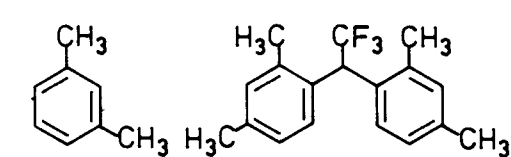
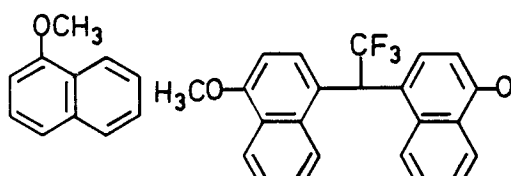
During our studies^{11,59} of the reaction of indole (or indoline) with $\text{NaBH}_4/\text{CF}_3\text{CO}_2\text{H}$, we observed the formation of an interesting bis-indole product (Eq. 26).



More recently, we have found¹³⁰ that this "Baeyer condensation"¹³¹ is general for activated arenes and generally furnishes the p,p'-isomer in fair to good yield (Table 23). The reaction fails with benzene, toluene, and p-xylene.

TABLE 23. Reaction of Arenes With $\text{NaBH}_4/\text{CF}_3\text{CO}_2\text{H}$ to Give 1,1,1-Trifluoro-2,2-diarylethanes

Entry	Substrate	Product	Conditions	Yield	Ref.
1			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ 60°	34%	59
2		"	NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ Δ	52%	59
3			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ Δ	22%	59
4		"	NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ Δ	46%	59
5			NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ Δ	47%	59
6	Ph_2O		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ Δ , 3 hr	53%	130

7		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ $\Delta, 3 \text{ hr}$	65%	130
8		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ $\Delta, 3 \text{ hr}$	48%	130
9		NaBH_4 $\text{CF}_3\text{CO}_2\text{H}$ 25	78%	130

XX. SUMMARY

In this review we have tried to illustrate the versatility of the relatively new acyloxyborohydride class of reducing agents. We have shown how, by changing carboxylic acid, solvent, stoichiometry, temperature, time, and hydride reagent itself, one can achieve remarkable chemoselectivity in an array of different types of reactions.

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